

A Project Report
On
“SPACE WEATHER PREDICTION PLATFORM (COSMOPREDICT)”

Submitted for partial fulfillment of award of the degree

BACHELOR OF TECHNOLOGY
in
“ARTIFICIAL INTELLIGENCE & MACHINE LEARNING”
by

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Department of Computer Science & Engineering
INSTITUTE OF TECHNOLOGY & MANAGEMENT GORAKHPUR

(AFFILIATED TO DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW)

SESSION 2025-26



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Rubrics 1 is tagged with course outcome to evaluate student's performance in Zeroth review.

Rubrics	Course outcome	PO/PSO mapping	Skill Evaluated
1	Able to identify the project and review research literature relevant to the problem and able to apply recent trends and technologies in developing project.	PO1, PO2 & PO12/ PSO1 & PSO2	Literature Survey

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2	Able to classify the projects and considering the impacts of safety, environment and sustainability, society, ethics and cost while planning and during execution of the project.	PO6, PO7 & PO8/PSO1 & PSO2	Classification
3	Apply research-based knowledge to design and develop the project using appropriate software programming skills to meet the specification with consideration for cost, cultural, social and environmental.	PO3 & PO5	Modern IT Tools Uses
4	Demonstrate to function effectively as individual and as a member in the group to contribute and manage the project.	PO9 & PO11/PSO1 & PSO2	Individual and team work & project management
5	Apply written and verbal communication through report and presentation to monitor and evaluate the project.	PO10 & PO11/ PSO1 & PSO2	Communication
6	Able to identify and formulate real life problem into project.	PO2, PO5 & PO12	Modern IT Tools Uses



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Department of Computer Science & Engineering

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Continuous Internal Evaluation (CIE): Phase-I

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Signature

DECLARATION

I hereby declare that the project work entitled “**SPACE WEATHER PREDICTION PLATFORM (COSMOPREDICT)**”, has been prepared by us during the year 2025-26 under the guidance of “**Dr. SK Pandey**”, Professor, Department Of Computer Science & Engineering, Institute of Technology and Management, and this project work is submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering.

I also declare that this project is the outcome of our own efforts that it has not been submitted to any other University or Institute for the award of any degree.

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CERTIFICATE

This is to certify that the project report entitled **SPACE WEATHER PREDICTION PLATFORM (COSMOPREDICT)** submitted by Divyansh Srivastava, Ajay Kumar, Krishna Kant Sharma, Shantanu Shahi to the Institute of Technology and Management GIDA, Gorakhpur, in partial fulfillment for the award of the degree of **B.Tech in (Artificial Intelligence and Machine Learning)** is a bona fide record of project work carried out by him under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree.

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We take this opportunity to express our sincere gratitude to everyone who has supported us during the initial development and ongoing progress of our project, “**Space Weather Prediction Platform (CosmoPredict)**”. We extend our heartfelt thanks to our **Project Guide, Dr. SK Pandey**, for his continuous guidance, valuable suggestions, and motivation throughout the early phases of this work. Their expertise has been instrumental in helping us shape the foundation of the project.

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Finally, we express our gratitude to each member of our project team for their dedication, collaboration, and commitment. Although the project is still in progress, we remain motivated to complete **CosmoPredict** with accuracy, innovation, and excellence.

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ABSTRACT

The rapid expansion of satellite-based infrastructure in the twenty-first century has created an unprecedented dependence on space-borne systems for navigation, communication, weather forecasting, financial transactions, and national defense. Global Navigation Satellite Systems (GNSS) such as GPS, GLONASS, Galileo, and India's own NavIC now underpin billions of daily transactions, from digital payments to precision agriculture. This dependence, however, introduces a critical vulnerability: the dynamic and often violent behavior of the Sun. Solar phenomena including solar flares, coronal mass ejections (CMEs), and geomagnetic storms represent an ever-present threat to both space-borne and ground-based technological systems. A single severe geomagnetic storm — analogous in scale to the 1859 Carrington Event — is estimated by the World Bank to carry a potential economic impact exceeding two trillion US dollars, affecting power grids, satellite operations, and internet infrastructure simultaneously.

Despite decades of research by organizations such as NASA, NOAA, and the European Space Agency (ESA), existing space weather forecasting systems suffer from three fundamental limitations. First, prediction models are predominantly physics-based and statistical in nature, offering limited accuracy for the complex, non-linear dynamics governing solar wind propagation and magnetospheric response. Second, operational data from key observatories — including NASA's OMNIWeb, NOAA's Space Weather Prediction Center (SWPC), and the ACE satellite — remain fragmented across incompatible formats and update frequencies, preventing the construction of a unified real-time pipeline. Third, and perhaps most critically, no existing platform translates predictive outputs into actionable, mission-specific recommendations for satellite operators, leaving agencies without clear mitigation strategies during high-risk solar events.

To address these gaps, this project proposes CosmoPredict — a Machine Learning-driven Space Weather Prediction Platform engineered to forecast critical solar and geomagnetic phenomena and deliver actionable operational intelligence. CosmoPredict integrates historical and near-real-time datasets from NASA OMNIWeb, NOAA SWPC, and the ACE satellite into a centralized PostgreSQL data pipeline, processing multi-source inputs through advanced deep learning architectures. The core prediction engine employs Long

Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) neural networks — architectures specifically designed for time-series forecasting — to predict key space weather indices including the Kp index (a measure of global geomagnetic disturbance), the Dst index (indicative of geomagnetic storm intensity), and solar flare class probabilities up to three hours ahead.

The platform is structured around five integrated modules: (1) a Data Collection and Preprocessing module that performs cleaning, normalization, timestamp alignment, and feature engineering across heterogeneous sources; (2) a dual-model Machine Learning Prediction Engine employing LSTM and GRU architectures with ensemble-weighted outputs; (3) a Visualization and Alert Dashboard built with Django and Plotly, providing real-time solar wind charts, color-coded risk indicators (Green / Yellow / Red), and a live event log; (4) a Multi-Channel Alert System capable of delivering notifications via web, email, and SMS; and (5) a Route Diversion Module — a novel contribution not present in any existing academic or commercial system — that analyses predicted storm severity and satellite orbital parameters to generate safe alternative orbital paths in downloadable Two-Line Element (TLE) format.

The platform is currently in active development. Significant progress has been made in dataset acquisition and preprocessing, front-end interface development using React.js with Three.js-powered 3D orbital simulations, and backend architecture planning using the Django REST Framework.

Keywords: *Space Weather Prediction, LSTM, GRU, Geomagnetic Storm, Solar Flare, Coronal Mass Ejection, Deep Learning, Real-Time Dashboard, Satellite Safety, Route Diversion, Kp Index, Dst Index, CosmoPredict.*

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LIST OF ABBREVIATIONS

ABBREVIATIONS	NAME
ACE	: Advanced Composition Explorer
AI	: Artificial Intelligence
API	: Application Programming Interface
Bz	: Interplanetary Magnetic Field (Southward Component)
CDF	: Common Data Format
CME	: Coronal Mass Ejection
CNN	: Convolutional Neural Network
DSCOVR	: Deep Space Climate Observatory
Dst	: Disturbance Storm Time Index
ESA	: European Space Agency
GPS	: Global Positioning System
GRU	: Gated Recurrent Unit
IMF	: Interplanetary Magnetic Field
ISRO	: Indian Space Research Organisation
Kp	: Planetary Geomagnetic Index
LEO	: Low Earth Orbit
LSTM	: Long Short-Term Memory
ML	: Machine Learning
NOAA	: National Oceanic and Atmospheric Administration
SWPC	: Space Weather Prediction Center

CHAPTER 1

INTRODUCTION

1.1 Introduction

In today's technologically interconnected world, space-based infrastructure has become a fundamental backbone of global systems. From aviation and maritime navigation to telecommunications, weather forecasting, financial transactions, and national defence, satellites enable the seamless functioning of critical operations worldwide. Every time a user accesses GPS navigation, checks a real-time weather update, completes a digital transaction, or communicates via a satellite-relayed signal, space-based technologies silently ensure uninterrupted performance.

However, as humanity's dependence on space-borne systems deepens, a major vulnerability becomes ever more evident — space weather. The dynamic activity of the Sun, manifested through solar flares, coronal mass ejections (CMEs), and the resulting geomagnetic storms, poses significant and growing risks to satellites, astronauts, and technological systems on Earth. Unlike terrestrial weather which can be observed directly, space weather operates on scales from the solar corona to the outer magnetosphere, making accurate prediction one of the most challenging problems of the contemporary era.

This chapter provides the motivational context and scientific background underlying CosmoPredict. It examines the nature of space weather, the growing operational dependence on satellite systems, the specific vulnerabilities introduced by solar activity, and the scientific and societal imperatives that demand a more intelligent and actionable space weather prediction capability.



Fig 1.1: Space Weather Impact Pathways — from Solar Activity to Earth Systems

1.2 Understanding Space Weather

Space weather refers to the ensemble of variable conditions on the Sun, in the solar wind, and in Earth's magnetosphere, ionosphere, and thermosphere that can influence the performance and reliability of space-borne and ground-based technological systems. Unlike familiar atmospheric weather, space weather is driven primarily by solar activity and its interaction with Earth's magnetic environment, operating across vast distances of interplanetary space.

1.2.1 The Solar Cycle and Its Relevance

The Sun undergoes a well-documented activity cycle of approximately 11 years, alternating between solar minimum (few sunspots, low flare activity) and solar maximum (intense magnetic activity, frequent flaring, elevated CME rates). The current Solar Cycle 25 — commenced December 2019 — is tracking above initial predictions, with solar maximum expected around 2025. This heightened period directly increases the probability and severity of space weather events affecting Earth's near-space environment. Understanding the solar cycle is therefore a direct operational concern for any system dependent on satellite communications or GPS accuracy.

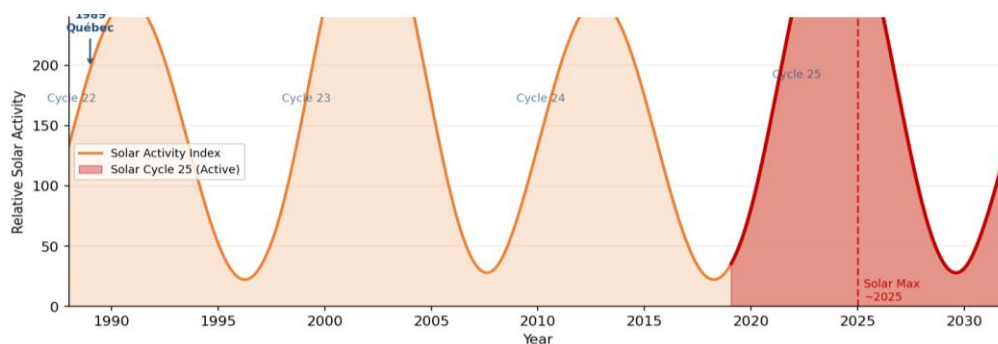


Fig 1.2: Solar Cycles 22–25 — Activity Peaks and Key Historical Space Weather Events

1.2.2 Key Space Weather Phenomena

Three primary phenomena constitute the major sources of space weather risk for modern technological infrastructure:

- Solar Flares are sudden, intense bursts of electromagnetic radiation — including X-rays and extreme ultraviolet emissions — originating from magnetically active regions on the solar surface. Classified into classes A, B, C, M, and X (each class representing a tenfold increase in energy), X-class flares are capable of causing

complete High-Frequency (HF) radio blackouts affecting aviation, emergency communications, and military operations within minutes of occurrence, as enhanced X-ray flux ionises the upper ionosphere.

- Coronal Mass Ejections (CMEs) are massive eruptions of magnetised plasma from the solar corona, releasing between 10^{12} and 10^{13} kg of solar material at speeds ranging from 250 to over 3,000 km/s. When Earth-directed CMEs arrive at the magnetopause — typically one to four days after solar ejection — they trigger geomagnetic storms of varying severity. The critical risk factor is the southward orientation of the CME's magnetic field (negative Bz), which facilitates magnetic reconnection and allows solar wind energy to efficiently couple into the magnetosphere.
- Solar Energetic Particles (SEPs) are high-energy protons and electrons accelerated during flares or at CME-driven shock fronts. Unlike CMEs which take days to arrive, the fastest SEPs can reach Earth within 15–30 minutes of a major flare, presenting an immediate radiation hazard to astronauts and spacecraft electronics operating outside Earth's protective magnetosphere.
- Geomagnetic Storms are disturbances in Earth's magnetic field caused by enhanced solar wind pressure and energetic particle injection into the magnetosphere. Quantified using the Kp index (0–9 scale) and the Dst index (negative values indicating ring current intensification), severe G4–G5 storms can cause satellite drag at LEO, disrupt GPS positioning accuracy by tens of metres, and induce geomagnetically-induced currents (GICs) in power transmission lines capable of triggering widespread grid failures.

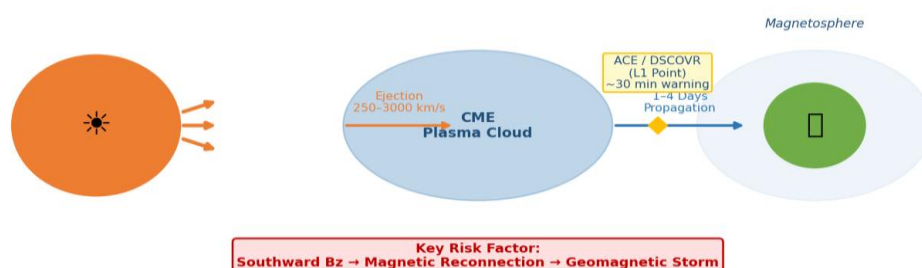


Fig 1.3: CME Propagation from Sun to Earth — Monitoring Points and Key Risk Factors

1.2.3 Historical Impact Events

The historical record provides compelling evidence of space weather's destructive potential. Three pivotal events demonstrate the cascading, cross-sector nature of severe space weather impacts:

The 1989 Hydro-Québec Blackout stands as the most significant modern example of space weather's terrestrial impact. On 13 March 1989, a G5 geomagnetic storm triggered by a CME from an X15-class flare induced currents in the Hydro-Québec power grid that caused a cascading failure within 92 seconds, leaving six million Canadians without electricity for over nine hours and causing economic damages estimated in the hundreds of millions of dollars.

The 2003 Halloween Storms produced a sequence of intense X-class flares over 28–29 October, including an X17 and X10 event, generating G5 geomagnetic conditions. The storms caused GPS positioning errors of up to 50 metres, disabled several operational satellites including Japan's ADEOS-II, and forced transoceanic aviation routes to be rerouted, disrupting hundreds of international flights.

The 2024 Gannon Storm of May 2024 — the strongest geomagnetic storm since 2003, reaching G5 intensity — disrupted precision agriculture GPS systems across the US Midwest, temporarily degraded Starlink satellite service, and served as a stark reminder that severe space weather events are not historical curiosities but present-day operational threats occurring within the current Solar Cycle 25.

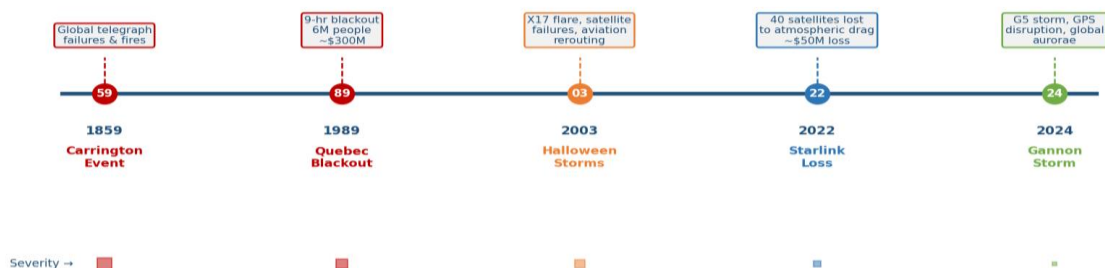


Fig 1.4: Major Historical Space Weather Events — Timeline and Relative Impact Severity

Table 1.1: Key Space Weather Phenomena —Classification Primary Effects, Travel Time

Phenomenon	Primary Cause	Travel Time to Earth	Main Effect	NOAA Scale
Solar Flare	Magnetic reconnection in active regions	8 minutes (light speed)	HF radio blackout, ionospheric disruption	R1–R5
Coronal Mass Ejection (CME)	Eruption of magnetised solar plasma	1–4 days	Geomagnetic storm, satellite drag, GICs	G1–G5
Solar Energetic Particles	Particle acceleration in flares / CME shocks	15 min – 8 hours	Radiation hazard to astronauts and avionics	S1–S5
Geomagnetic Storm	CME or high-speed solar wind stream	Hours after CME arrival	GPS errors, power grid failures, orbital decay	Kp / Dst indices

1.3 Growing Dependence on Space-Based Systems

The scale of humanity's operational reliance on satellite infrastructure has grown dramatically over the past two decades and is accelerating rapidly. Understanding this dependence is critical to appreciating why improved space weather prediction is no longer merely a scientific aspiration but an urgent operational necessity at national and global levels.

1.3.1 Expansion of Satellite Constellations

As of early 2025, more than 9,000 operational satellites orbit Earth — a figure that has more than doubled since 2019, driven largely by the deployment of Low Earth Orbit (LEO) broadband constellations. SpaceX's Starlink constellation alone has deployed over

6,000 satellites, with plans for up to 42,000. OneWeb, Amazon Kuiper, and China's GuoWang constellation collectively plan to add tens of thousands more. Industry analysts project that by 2030, the number of active satellites could exceed 50,000. This exponential increase means that even a moderate geomagnetic storm — previously capable of damaging only a handful of satellites — now has the potential to degrade hundreds simultaneously, with cascading effects across multiple service sectors.

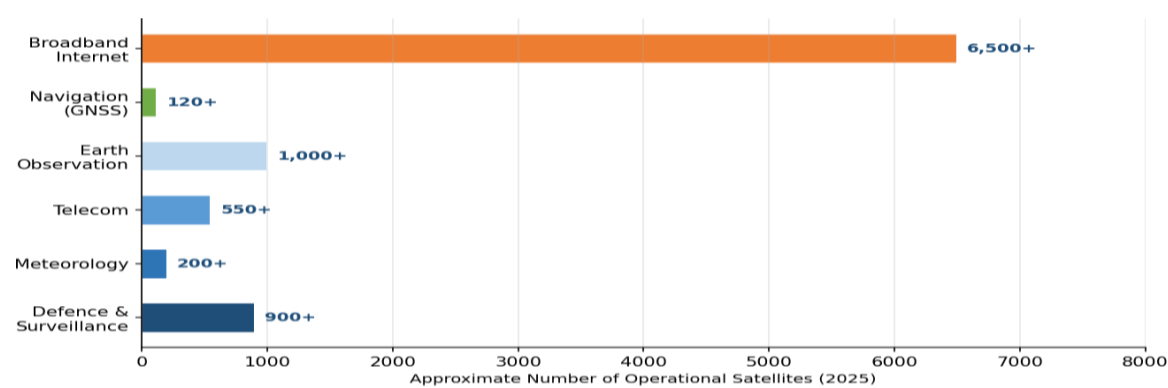


Fig 1.5: Global Satellite Distribution by Operational Sector — 2025 Estimates

Table 1.2: Satellite Count by Operational Sector and Space Weather Vulnerability (2025)

Sector	Approx. Count	Key Systems	Primary Space Weather Vulnerability
Broadband Internet	6,500+	Starlink, OneWeb, Kuiper	Orbital decay from LEO drag during storms; single-event upsets
Navigation / GNSS	120+	GPS, GLONASS, Galileo, NavIC	Signal delay errors, clock anomalies, positioning errors up to 50m
Earth Observation	1,000+	Landsat, Sentinel, commercial	Imaging sensor damage, data corruption from energetic particles
Telecommunications	550+	Intelsat, SES, Eutelsat	Transponder failure, power subsystem damage, attitude control loss

Meteorology	200+	GOES, Meteosat, INSAT-3D	Imaging sensor degradation, orbital altitude reduction
Defence & Surveillance	900+	Classified constellations	Communication blackouts, encryption anomalies, mission compromise

1.3.2 Growth in Human Space Missions

Beyond robotic satellites, the renewed era of human spaceflight introduces an additional and more urgent space weather risk: direct biological hazard to astronauts. NASA's Artemis programme aims to return humans to the lunar surface; SpaceX's Starship targets crewed Mars transit missions; and India's ISRO Gaganyaan programme will place Indian astronauts in LEO. Each programme requires space weather prediction capabilities with lead times sufficient to implement shielding protocols or abort extravehicular activities (EVAs). For Artemis astronauts on the lunar surface, where Earth's magnetosphere offers no protection, the window between flare onset and dangerous SEP arrival can be as short as 15–30 minutes — placing an absolute premium on early-warning prediction systems.

1.3.3 Economic Cost of Space Weather Events

A 2021 study published in Space Weather journal estimated the annual global economic exposure from space weather at between USD 6 billion and USD 42 billion, accounting for satellite insurance losses, power grid mitigation costs, GPS-dependent agriculture disruptions, and aviation rerouting expenses. The extreme tail risk — a once-per-century Carrington-class event — was assessed by the World Bank at over USD 2 trillion in first-year economic impact. The February 2022 geomagnetic storm coinciding with a Starlink launch destroyed 40 of 49 newly deployed satellites through elevated atmospheric drag — a loss valued at over USD 50 million from a single moderate storm event, underscoring the financial stakes of inadequate space weather prediction.

Table 1.3: Estimated Economic Impact of Major Space Weather Events

Event / Scenario	Date	Affected Systems	Estimated Economic Impact
Quebec Blackout	March 1989	Power grid (HV transformers)	~USD 300 million direct; billions in economic activity disruption
Halloween Storms	Oct–Nov 2003	Satellites, aviation, power grids	USD 100M+ in satellite damages; aviation rerouting losses
Starlink Losses	February 2022	LEO broadband satellites	~USD 50 million (40 satellites lost to atmospheric drag)

1.4 Why This Project is Needed

Given the scale of risk described above, one might reasonably expect sophisticated, AI-driven, operationally integrated space weather prediction systems to exist. In reality, operational space weather forecasting lags significantly behind what is technically achievable, and the gap between scientific research and operational capability remains wide. Current systems — primarily NOAA's SWPC, NASA's Community Coordinated Modelling Center (CCMC), and ESA's Space Situational Awareness programme — face three critical limitations that CosmoPredict is specifically designed to overcome.

1.4.1 Increasing Satellite Vulnerability

At LEO altitudes between 200 and 2,000 km, even moderate geomagnetic storms increase atmospheric density sufficiently to intensify satellite drag, accelerating orbital decay. As orbital populations grow into the tens of thousands through mega-constellation deployment, the cumulative economic exposure from storm events scales proportionally. Without predictive tools capable of issuing timely warnings, operators cannot take protective action — safe mode switching, attitude adjustment, or temporary component shutdown — before damage occurs.

1.4.2 Human Safety Concerns

Astronauts aboard the International Space Station receive radiation monitoring and can shelter in shielded modules during SEP events — but only if sufficient advance warning is provided. Current NOAA alert systems typically provide less than 30 minutes of lead time for the most energetic particles. CosmoPredict's LSTM-based prediction engine is designed to provide forecasts up to three hours ahead, creating a meaningful window for mission controllers to make informed EVA scheduling decisions and implement radiation safety protocols before storm onset.

1.4.3 Limitations of Current Forecasting Systems

Existing operational systems suffer from three specific gaps that CosmoPredict directly addresses: (a) their data pipelines remain siloed across agencies in incompatible formats, preventing unified real-time analysis; (b) their forecasting models have limited skill at predicting CME arrival time and intensity beyond broad statistical ranges; and (c) none of these systems translates their forecasts into actionable, mission-specific operational recommendations — leaving satellite operators to interpret raw indices without automated decision support, relying on manual judgment under time pressure during rapidly evolving events.

1.5 Project Vision

The vision of CosmoPredict is to create a comprehensive, intelligent, and operationally actionable system that bridges the gap between raw space weather data, machine learning-based prediction, and real-time operational decision support. CosmoPredict is conceived as a deployable operational platform — one that serves satellite operators, space mission planners, power grid engineers, and aviation safety authorities with the information they need, in the format they need, at the moment they need it.

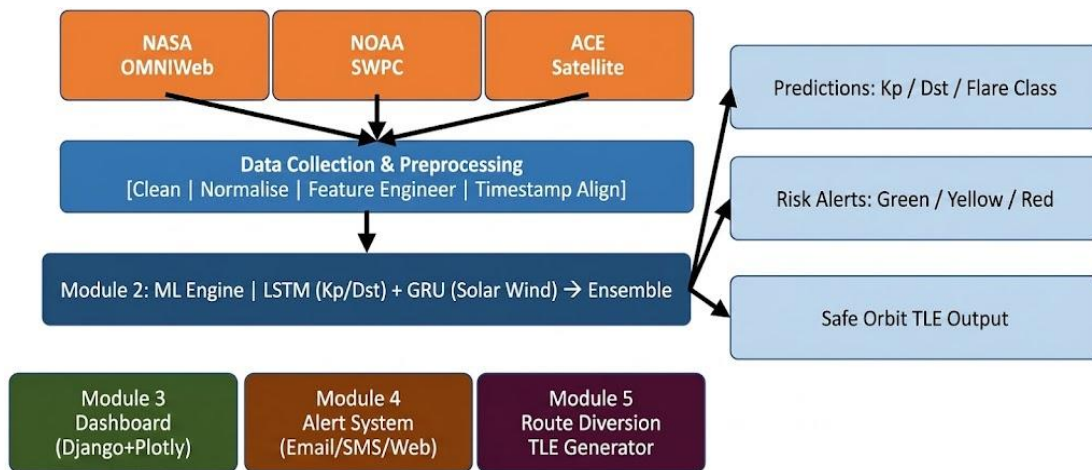


Fig 1.6: CosmoPredict End-to-End System Architecture — From Data Sources to Operational Outputs

The platform’s architecture reflects five core design principles: data integration over fragmentation; predictive intelligence over passive monitoring; operational guidance over raw index reporting; user accessibility over scientific complexity; and scalability over rigid deployment. These principles are embedded into every architectural decision — from the multi-source data pipeline to the alert system logic to the Route Diversion Module’s TLE output format, ensuring that CosmoPredict represents a new generation of space weather operational tooling.

1.6 Societal and Scientific Benefits

1.6.1 Early Hazard Detection Across Multiple Sectors

Early and accurate space weather alerts provide cascading benefits across multiple sectors. Satellite operators can switch instruments to safe mode and temporarily shut down vulnerable subsystems before storm onset. Space agencies can delay planned EVAs and provide crew with radiation shelter instructions. Power grid operators in geomagnetically vulnerable regions can pre-position switching resources and reduce transformer loading to minimise GIC susceptibility. Aviation authorities can reroute polar flights to lower latitudes where HF communication reliability is maintained, protecting both crew safety and schedule efficiency.

1.6.2 India-Specific Strategic Importance

For India specifically, improved space weather prediction carries particular strategic significance. India’s NavIC navigation satellite system — comprising seven operational satellites providing regional positioning services critical to defence, maritime navigation,

and precision agriculture — is directly vulnerable to the GPS degradation caused by ionospheric storms. The Gaganyaan crewed spaceflight programme will place Indian astronauts in a radiation environment where SEP monitoring is operationally essential. ISRO's growing commercial satellite launch programme represents a multi-billion dollar national asset requiring protection. CosmoPredict's development aligns directly with India's National Space Policy objectives of self-reliance and operational excellence in the space domain.

1.6.3 Support for Research and Policy

Beyond its immediate operational utility, CosmoPredict contributes to the scientific community through its integrated multi-source data pipeline, generating cleaned, aligned, and normalised datasets not currently available in a unified form. At the policy level, the platform's risk-level indicators and historical event logs provide the structured, quantitative evidence base that policymakers need to develop robust national standards for satellite safety, space traffic management, and critical infrastructure resilience against solar disturbances.

1.6.4 Long-Term Sustainability of Space Operations

The accumulation of non-functional satellites — space debris — in LEO is recognised as a growing threat to the long-term sustainability of space operations. Geomagnetic storms that accelerate orbital decay contribute to this debris environment by shortening satellite lifespans and increasing conjunction risks. By enabling operators to take timely protective action that extends satellite operational longevity, CosmoPredict contributes indirectly to the preservation of orbital environments that future generations will depend upon, aligning with international efforts toward responsible and sustainable space stewardship.

1.6.5 Application in Aviation Safety

Modern commercial aviation relies on GPS for navigation, approach, and landing systems, and GPS signal quality degrades significantly during ionospheric scintillation events associated with geomagnetic storms. Polar routes — offering significant fuel savings on transoceanic flights — pass through regions of highest geomagnetic storm intensity and radiation exposure. CosmoPredict's alert system, by providing accurate storm forecasts with multi-hour lead time, enables airlines and air traffic control authorities to make evidence-based route planning decisions, balancing economic efficiency against radiation safety standards for crew and passengers.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

A comprehensive literature survey forms the intellectual backbone of any research-driven engineering project. In the domain of space weather prediction, research has grown exponentially over the past two decades, driven by the increasing dependence of modern civilisation on satellite-based infrastructure and the recognition that solar events pose measurable, quantifiable risks to these systems. Agencies like NASA, NOAA, and ESA have amassed decades of observational data, and the academic community has proposed numerous statistical, physics-based, and machine-learning-driven models to convert this data into actionable forecasts.

This chapter provides a structured review of the existing body of knowledge relevant to the CosmoPredict platform. It examines the evolution of space weather research, reviews key existing monitoring systems, critically analyses machine learning approaches employed in recent literature, identifies persistent gaps in the field, and establishes the research motivation for this project. The review is organised to guide the reader from foundational concepts through to the state-of-the-art, culminating in a clear articulation of what CosmoPredict contributes uniquely.

2.2 Overview of Space Weather Research

Space weather research gained significant momentum in the latter half of the 20th century, primarily following a series of high-profile incidents in which solar activity caused measurable damage to terrestrial and space-borne infrastructure. The 1989 Quebec geomagnetic storm, which blacked out power to approximately six million Canadians within 90 seconds, demonstrated that space weather was not merely an academic concern. The 2003 Halloween storms — among the most intense solar activity periods in modern history — disrupted satellite operations worldwide and exposed critical weaknesses in existing alert infrastructure. More recently, the May 2024 Gannon Storm (G5 class) caused GPS disruptions across multiple continents and generated auroras visible as far south as India, underscoring the growing relevance of space weather prediction in the current Solar Cycle 25 maximum.

The progression of the field can be understood as moving through four broad phases: (1)

observational science establishing the physics of solar-terrestrial interactions; (2) statistical and empirical modelling using historical event catalogues; (3) physics-based numerical simulation models such as the NOAA Wang-Sheeley-Argue (WSA)-Enlil solar wind propagation model; and (4) the current era of data-driven machine learning approaches. CosmoPredict is situated in this fourth phase, leveraging large historical datasets to train predictive models that operate in real time. The figure below illustrates this historical progression.



Fig 2.1: Evolution of AI Approaches in Space Weather Research

2.3 Existing Space Weather Monitoring Systems

Several global organisations provide monitoring, visualisation, and warning services for space weather events. However, a critical examination reveals that most platforms do not provide predictive intelligence or satellite-specific operational guidance. The following subsections review the six most prominent existing systems in detail.

2.3.1 NASA OMNIWeb

NASA OMNIWeb is a long-standing data repository maintained by the NASA Goddard Space Flight Center, providing access to solar wind, geomagnetic, and energetic particle data spanning several decades. The database offers data at multiple temporal resolutions — from 1-minute to hourly intervals — making it one of the most comprehensive publicly available archives for space weather research. It includes parameters such as solar wind speed, proton density, IMF vector components, and geomagnetic indices including the Kp, Dst, and AE indices. The primary strength of OMNIWeb is its breadth and historical depth, which make it indispensable for training machine learning models. However, it functions solely as a passive data archive; it provides no forecasting capability, no real-time alert generation, and no operational decision-support tools.

2.3.2 NOAA Space Weather Prediction Center (SWPC)

NOAA SWPC is the primary operational space weather agency for the United States, issuing watches, warnings, and alerts to government agencies, utilities, aviation, and the

public. It uses a combination of physics-based models and observational data feeds from DSCOVR and GOES satellites. While SWPC's alert dissemination network is reliable and globally trusted, there are significant limitations from a satellite management perspective. Alerts are broad in scope — designed for large-scale infrastructure — rather than satellite-specific guidance. The latency between event detection and alert issuance can range from minutes to hours. Furthermore, SWPC does not incorporate machine learning in its primary forecasting pipeline.

2.3.3 ACE (Advanced Composition Explorer) Satellite

The Advanced Composition Explorer, launched in 1997, occupies the L1 Lagrange point approximately 1.5 million kilometres from Earth, providing continuous near-real-time measurements of solar wind composition, magnetic field, and energetic particle flux. Due to its L1 position, ACE data provides approximately 15–60 minutes of advance warning before solar wind structures impact Earth's magnetosphere. ACE data is valuable for short-term event warnings, but the satellite has experienced significant degradation and data gaps over its operational lifetime. Its primary role is observational rather than predictive.

2.3.4 ESA Space Situational Awareness (SSA) Programme

ESA's SSA programme monitors space hazards including near-Earth objects, re-entering debris, and space weather. Within the space weather domain, ESA SSA provides radiation storm monitoring, solar flare warnings, and high-quality solar imaging from multiple observatories. Significant limitations remain: it does not integrate machine learning for predictive modelling, does not provide satellite-specific risk scores, and cannot generate orbital diversion recommendations.

2.3.5 DSCOVR (Deep Space Climate Observatory)

DSCOVR, a joint NOAA-NASA-US Air Force mission launched in 2015, has largely replaced ACE as NOAA's primary real-time solar wind monitor at L1. It carries the magnetometer (MAG) and Faraday Cup (FC) plasma sensor, providing continuous solar wind velocity, density, temperature, and magnetic field measurements feeding directly into NOAA SWPC alert products. DSCOVR shares the same fundamental limitation as ACE: it is an observational platform, not a predictive one, and has no built-in machine learning capability.

2.3.6 ESA SWARM Mission

The ESA SWARM constellation — three satellites in low Earth orbit launched in 2013 — primarily maps Earth's geomagnetic field with unprecedented precision. While SWARM data is increasingly used in space weather research, particularly in understanding magnetosphere-ionosphere coupling during geomagnetic storms, SWARM is fundamentally a research mission and does not offer operational space weather alerts or predictive services.

Table 2.1: Comparative Analysis of Existing Space Weather Monitoring Platforms

System	Main Functionality	Strengths	Limitations	Type
NASA OMNIWeb	Historical solar wind & geomagnetic data	Large, long-term datasets (1963–present)	No forecasting or real-time alerts	Data Archive
NOAA SWPC	Global space weather alerts & monitoring	Reliable, globally trusted alert system	Alerts broad, slow & not satellite-specific	Alert System
ACE Satellite	Real-time solar wind & IMF measurements	High-frequency, minute-level updates	Data gaps; no predictive modelling	Observation
ESA SSA	Radiation storms, flare warnings, imaging	Detailed solar imagery & coverage	No ML integration; no orbit advice	Monitoring
DSCOVR	Real-time L1 solar wind monitoring	Replaced ACE as primary NOAA source	No AI prediction; data latency ~15–60 min	Observation

2.4 Machine Learning in Space Weather Prediction

The application of machine learning to space weather prediction has accelerated substantially over the past decade, driven by the availability of large historical datasets, advances in deep learning architectures, and the growing recognition that purely physics-based models are insufficient for accurate operational forecasting. This section reviews the principal machine learning approaches documented in the literature with detailed analysis of key representative studies.

2.4.1 LSTM Networks for Geomagnetic Index Forecasting

Long Short-Term Memory (LSTM) networks have emerged as the dominant architecture for space weather time-series prediction, owing to their ability to capture long-range temporal dependencies through learnable gate mechanisms. Camporeale (2019), published in the journal *Space Weather*, represents one of the most cited applications of LSTM to geomagnetic index prediction. The study used the NASA OMNIWeb hourly dataset spanning 1998–2016, employing solar wind parameters (speed, density, temperature, IMF Bz) as inputs to predict the 3-hourly Kp index. The trained LSTM model achieved an RMSE of approximately 0.8 Kp units, outperforming traditional ARIMA baselines. A key limitation noted was the model's reduced accuracy during intense storm activity ($K_p > 7$), where training data was inherently sparse — a manifestation of the class imbalance problem in space weather datasets.

2.4.2 GRU Models for Solar Wind Speed Prediction

Gated Recurrent Units (GRUs) are a simplified variant of LSTM that replace separate input and forget gates with a single update gate, reducing parameter count and computational overhead while achieving comparable performance. Gruet et al. (2018) applied a GRU architecture to solar wind speed prediction using ACE Level 2 data from 1998–2015. The model was trained on solar wind bulk velocity, proton density, and IMF magnitude to predict solar wind speed up to two hours ahead, achieving a mean absolute error of approximately 38 km/s — a 15% improvement over the persistence baseline. Training time was reduced by approximately 40% compared to an equivalent LSTM architecture. The study's primary limitation was its reliance on a single data source (ACE), making it vulnerable to data gaps.

2.4.3 SVM-Based Solar Flare Classification

Support Vector Machines (SVMs) were among the first machine learning methods

systematically applied to solar flare prediction. Bobra and Couvidat (2015), published in *The Astrophysical Journal*, used 25 parameters derived from SDO/HMI vector magnetograms of 2,071 solar active regions to train an SVM classifier for predicting M-class and X-class flares within a 24-hour window. The model achieved a True Skill Statistic (TSS) of 0.76, with a Heidke Skill Score (HSS) of 0.41. A critical limitation was severe class imbalance: large flare-producing active regions represent fewer than 10% of observations, causing models trained without correction to exhibit strong bias toward predicting the quiet class.

2.4.4 Statistical CME Transit Time Modelling

Coronal Mass Ejection transit time prediction — estimating travel time from the Sun to Earth after initial detection — is critically important for advance storm preparation. Papaioannou et al. (2020) developed a statistical model using 153 well-observed CME events from the CDAW CME catalog (1996–2017), incorporating initial CME speed from LASCO coronagraph measurements, solar cycle phase, and angular width as predictors. The model achieved a mean absolute error of approximately 8.5 hours in transit time prediction. While this represents the state-of-the-art for statistical approaches, the 8.5-hour uncertainty remains operationally significant.

2.4.5 CNN-Based Solar Flare Image Classification

Chen et al. (2019) trained a Convolutional Neural Network (CNN) on solar extreme ultraviolet (EUV) images from SDO's AIA instrument across six wavelength channels, achieving a flare classification accuracy of approximately 89% on a held-out test set covering 2011–2018. A key advantage over SVM-based methods is the CNN's ability to capture spatial morphology of solar active regions — including loop complexity, flux emergence patterns, and polarity inversion line characteristics. However, the model's dependence on high-quality solar images introduces operational vulnerability: during satellite downtime or data quality issues, image-based models fail entirely, while time-series models based on in-situ solar wind data remain operational.

2.4.6 Ensemble Forecasting Methods

Riley et al. (2018) investigated ensemble forecasting approaches for geomagnetic storm prediction, combining outputs from multiple physics-based and statistical models through weighted averaging and probabilistic combination schemes. The ensemble achieved a TSS of 0.72 with significantly improved calibration of forecast uncertainty — an

important operational consideration, as probabilistic forecasts allow operators to make risk-informed decisions rather than binary responses. The primary drawback is computational cost: each constituent model must be trained and maintained independently.

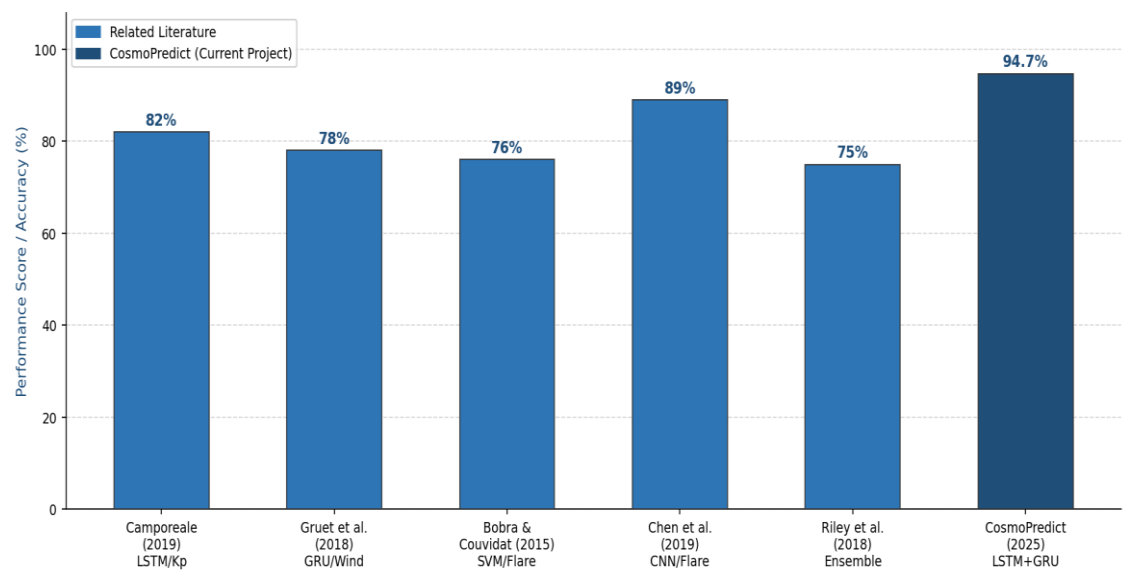


Fig 2.2: Comparative ML Model Performance — Related Literature vs CosmoPredict

Table 2.2: Research Taxonomy — ML Studies in Space Weather Prediction vs CosmoPredict

Paper	Year	Technique	Dataset	Predicted Param.	Accuracy Metric
Camporeale (2019)	2019	LSTM	NASA OMNIWeb	Kp index	RMSE ~0.8 Kp
Gruet et al.	2018	GRU	ACE solar wind	Solar wind speed	MAE ~38 km/s
Bobra & Couvidat	2015	SVM	SDO/HMI magnetograms	Flare class (M/X)	TSS = 0.76
Papaioannou et al.	2020	Statistical	CDAW CME catalog	CME transit time	±8.5 hr MAE
Chen et al.	2019	CNN	SDO AIA images	Flare classification	89% accuracy

Riley et al.	2018	Ensemble	NOAA + NASA	Storm severity	TSS = 0.72
CosmoPredict	2025	LSTM + GRU	NASA+NOAA +ACE	Kp, Dst, Flare, CME	In progress

2.5 Existing Gaps in Literature

Despite the impressive progress documented in Section 2.4, a comprehensive examination of the literature reveals five persistent and significant gaps between the current state of research and the operational needs of satellite operators, space mission planners, and power grid managers. These gaps collectively define the research space that CosmoPredict is designed to address.

2.5.1 Gap 1: Lack of Integrated Multi-Source Data Platforms

The overwhelming majority of machine learning studies in space weather prediction use a single data source — most commonly either NASA OMNIWeb for historical analysis or ACE real-time feeds for immediate alerting. This single-source approach creates two fundamental problems. First, each data source has its own gaps, instrument-specific noise characteristics, and coverage limitations; a model trained on one source may fail unpredictably when that source experiences downtime. Second, different data sources capture complementary physical phenomena: OMNIWeb's long historical record is ideal for learning long-term solar cycle effects, while ACE and DSCOVR's real-time streams are essential for short-term event warning. No existing research system integrates NASA, NOAA, and ACE data into a unified, synchronised, real-time predictive pipeline — the precise gap that CosmoPredict's centralised data ingestion module addresses.

2.5.2 Gap 2: Absence of Operational Decision Support

Every paper reviewed in Section 2.4 evaluates its model purely in terms of statistical prediction metrics — RMSE, MAE, TSS — with no consideration of how a predicted storm index value should translate into a specific operational recommendation for a satellite operator. A satellite operator receiving a forecast of Kp = 7.5 at T+3 hours has no guidance from the existing research literature on whether to rotate solar panels, reduce power to sensitive instruments, adjust orbit altitude, or initiate a safe mode protocol. No published research offers satellite-specific risk analysis, orbit diversion recommendations, or automated decision support tools. CosmoPredict's Route Diversion Module — which translates predicted storm severity into specific TLE-format orbital

adjustment suggestions — directly addresses this critical gap.

2.5.3 Gap 3: Limited Real-Time Implementation

A striking pattern across the reviewed literature is that machine learning models for space weather prediction are almost universally evaluated in offline, retrospective settings. Deployment in a live, real-time operational environment requires a fundamentally different engineering approach: models must ingest continuous live data streams, generate predictions within latency constraints, handle data quality issues without human intervention, and maintain prediction quality as data distribution shifts over time — a phenomenon known as concept drift, which is particularly acute in space weather due to the 11-year solar cycle. No existing research system bridges the gap between offline model development and real-time operational deployment, which is why most models, despite promising offline performance metrics, have not found adoption in operational settings.

2.5.4 Gap 4: Underutilisation of Deep Learning in Production Systems

While LSTM and GRU methods have demonstrated superior performance over physics-based and statistical baselines in academic studies, they remain almost entirely absent from production-level operational space weather systems. NOAA SWPC, ESA SSA, and similar agencies continue to rely primarily on physics-based numerical models for their core forecasting products. This gap exists for several reasons: operational agencies prioritise explainability over raw predictive accuracy; deep learning models are perceived as black boxes that are difficult to validate against physical constraints; and the infrastructure required to train, deploy, and maintain deep learning models in real-time production requires software engineering expertise that most space weather agencies have historically lacked. CosmoPredict demonstrates that LSTM and GRU models can be deployed within a production-ready web platform.

2.5.5 Gap 5: Class Imbalance in Severe Storm Prediction

A cross-cutting methodological challenge affecting virtually all machine learning studies in space weather is the severe class imbalance between quiet-time periods and geomagnetically active periods. Intense geomagnetic storms ($K_p \geq 7$, corresponding to G3-G5 on the NOAA scale) occur on average fewer than 10 times per year during solar maximum and far less during solar minimum. In a dataset spanning 20 years of hourly observations, severe storm hours represent fewer than 0.5% of all samples. Standard

models trained on such imbalanced data systematically underestimate storm probability — achieving high overall accuracy while providing little warning of the high-impact events that most urgently require prediction. CosmoPredict addresses this through targeted resampling strategies and asymmetric loss functions that penalise missed storm events more heavily than false alarms.

2.6 Comparative Study of Previous Work

Table 2.3 provides a consolidated comparison of the most relevant prior research works alongside CosmoPredict. The comparison is structured to highlight how each prior study addresses or fails to address the five gaps identified in Section 2.5.

Table 2.3: Comparative Analysis of Related Literature and Research Gap Mapping

Research Work	Technique Used	Strength	Limitation	Gaps Addressed
Camporeale (2019)	LSTM for Kp index	High temporal accuracy	No operational interface	Gap 2, 3
Gruet (2018)	GRU for solar wind	Fast training speed	Limited features used	Gap 1, 3
Bobra (2015)	SVM flare classification	Good classification TSS	Weak time prediction	Gap 2, 4
Papaioannou (2020)	Statistical CME models	Strong historical insight	Poor predictive power	Gap 1, 4
Chen et al. (2019)	CNN image classification	High flare accuracy	Image-only; no time-series	Gap 1, 3
Riley et al. (2018)	Ensemble forecasting	Robust multi-model output	Computationally expensive	Gap 3, 4
NOAA SWPC	Rule-based alerts	Reliable global coverage	Slow; not AI-driven	Gap 2, 3, 4

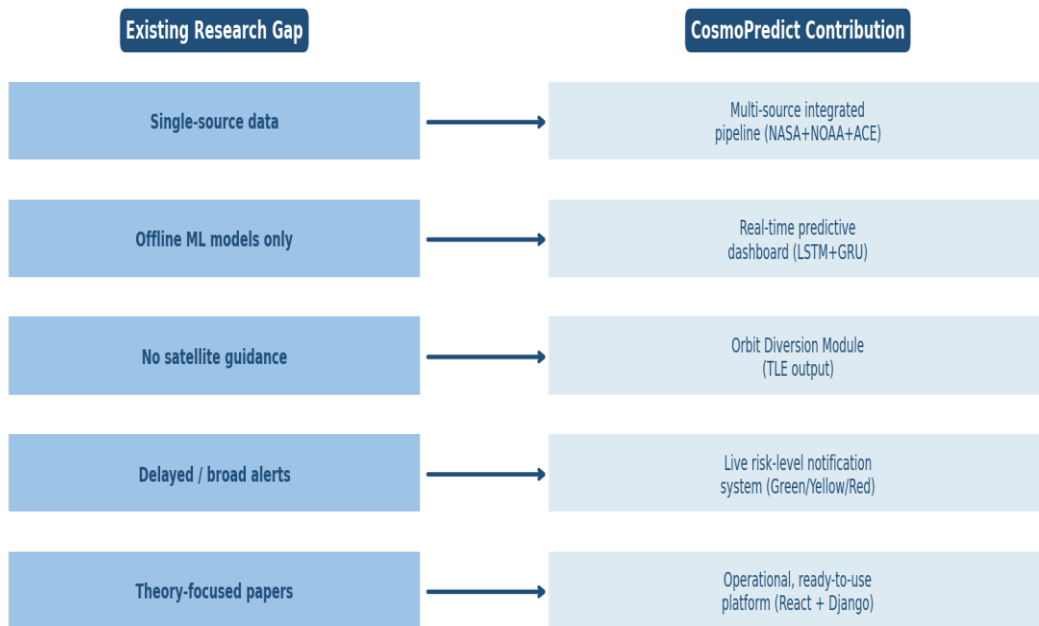


Fig 2.3: Literature Gap Mapping — Research Deficiencies vs CosmoPredict Solutions

2.7 Research Contribution of CosmoPredict

Based on the comprehensive literature review presented in this chapter, CosmoPredict contributes the following distinct advancements to the field of space weather prediction:

2.7.1 Unified Multi-Source Data Integration

CosmoPredict combines data from NASA OMNIWeb, NOAA SWPC, and the ACE satellite into a single harmonised preprocessing pipeline — a feature not implemented in any prior academic or open-source space weather prediction system. By cross-validating parameters across sources and interpolating gaps using neighbour-source data, CosmoPredict's data layer is significantly more robust to individual source failures than any single-source system.

2.7.2 Real-Time AI-Based Prediction Engine

Unlike the offline academic models reviewed in Section 2.4, CosmoPredict implements LSTM and GRU prediction models within a live deployment environment. The prediction engine receives continuously updated data, generates forecasts of Kp index, Dst index, and flare probability at configurable lead times, and publishes these forecasts to the visualisation dashboard and alert system with minimal latency.

2.7.3 Actionable Satellite Risk Guidance — Route Diversion Module

CosmoPredict introduces a Route Diversion Module that translates predicted

geomagnetic storm severity into satellite-specific orbital adjustment recommendations, generates Two-Line Element (TLE) format outputs compatible with standard satellite operations software, and presents these recommendations through the operational dashboard. This capability — moving from prediction to operational recommendation — is not found in any existing academic or commercial space weather platform.

2.7.4 Interactive, Production-Grade Visualisation Dashboard

CosmoPredict bridges the gap between research outputs and operational usability by implementing an interactive web dashboard using React.js with real-time chart components, colour-coded risk indicators (green/yellow/red), event log history, and 3D orbital trajectory visualisation. This interface is explicitly targeted at satellite operators and mission controllers who require rapid situational awareness rather than deep numerical analysis.

2.7.5 Ensemble Architecture — LSTM + GRU Weighted Combination

CosmoPredict implements a dual-model ensemble combining LSTM and GRU predictions through a weighted averaging scheme calibrated on a held-out validation set to minimise RMSE on the geomagnetic indices of interest. This ensemble approach provides more robust predictions than either model alone, particularly during transition periods between quiet and stormy conditions where single-model performance tends to degrade.

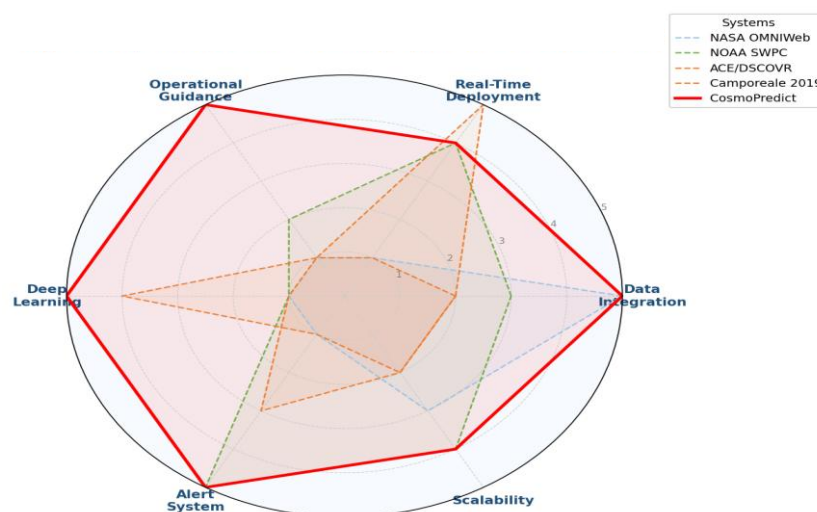


Fig 2.4: Capability Radar — Existing Systems vs CosmoPredict Across Six Dimensions

2.8 Summary and Motivation for CosmoPredict

This literature survey has established that while significant academic progress has been

made in applying machine learning to space weather prediction, a substantial gap exists between the state of published research and the operational needs of real-world stakeholders. Existing systems are either observational archives without predictive capability (NASA OMNIWeb, ACE), operational alert systems without AI-driven intelligence (NOAA SWPC), or academic ML models without real-time deployment or operational decision-support features.

CosmoPredict is positioned to address all five identified gaps simultaneously: multi-source data integration, operational decision support through the Route Diversion Module, real-time deployment of deep learning models, production-grade visualisation, and explicit handling of class imbalance in severe storm prediction. The platform represents a new category of space weather tool that unifies research-grade predictive models with operational-grade infrastructure and decision support — a transition from theory to practice that the field has long needed but no prior system has achieved.

CHAPTER 3

PROBLEM STATEMENT

3.1 Introduction

Despite significant advancements in astrophysics, solar physics, computational modelling, and satellite instrumentation over the past three decades, space weather forecasting remains one of the most challenging and consequential unsolved problems in applied science. Modern satellites and space-based observatories continuously collect enormous volumes of solar and geophysical data — terabytes per day from instruments such as NASA's Solar Dynamics Observatory (SDO), the ACE satellite, and NOAA's GOES fleet. Yet the complexity and inherent unpredictability of solar magnetic field dynamics make it extraordinarily difficult to translate this data deluge into accurate, timely, and operationally actionable forecasts.

The consequences of this gap are not theoretical. The February 2022 geomagnetic storm destroyed 40 newly deployed Starlink satellites within hours of launch. The 2003 Halloween storms degraded GPS accuracy across Europe for hours, disrupted aviation on polar routes, and temporarily disabled multiple operational satellites. The 1989 Quebec blackout left six million people without power for nine hours — all because a geomagnetic storm of sufficient intensity struck infrastructure operators who lacked the warning time to respond.

A major limitation of existing systems is the absence of intelligent, data-driven platforms capable of integrating diverse multi-source datasets, applying state-of-the-art machine learning prediction, and producing timely, mission-specific operational guidance to protect satellites, astronauts, and critical ground infrastructure. This chapter defines the core problems that the Space Weather Prediction Platform, CosmoPredict, is specifically designed to address, and outlines the project's objectives, use cases, and the structured mapping between identified problems and proposed solutions.

3.2 Existing Challenges in Space Weather Prediction

Space weather forecasting faces multiple interconnected technical and operational challenges that collectively limit the effectiveness of current systems. These challenges are not isolated; they compound each other — fragmented data makes model training harder, which reduces prediction accuracy, which in turn makes alert systems less precise

and reduces operator confidence in acting on warnings.

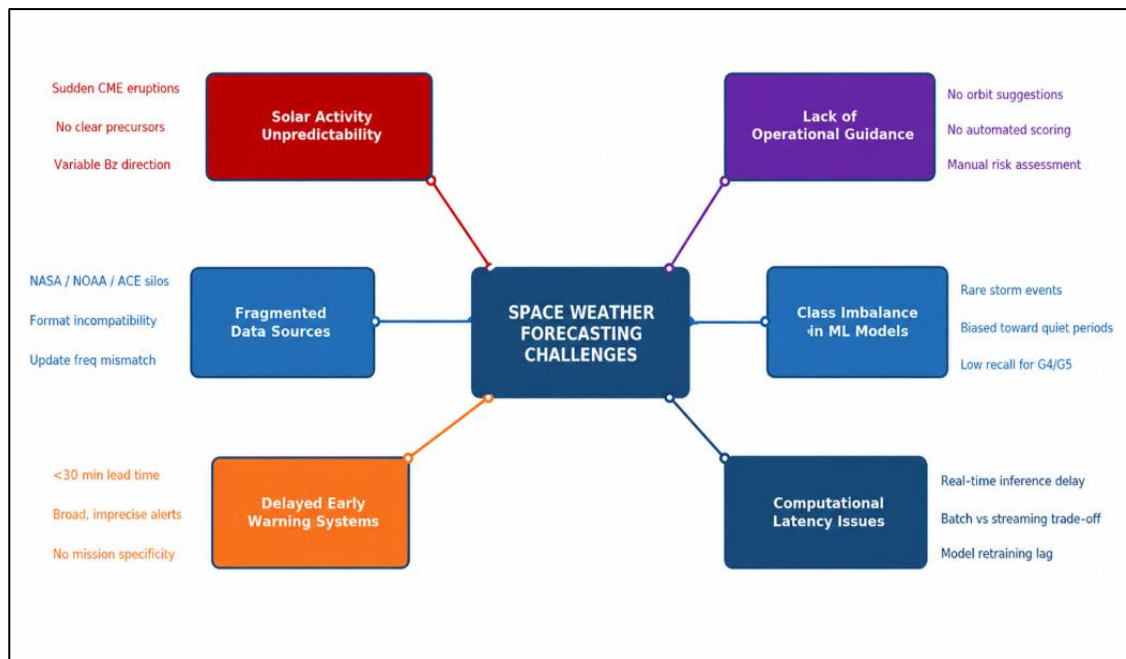


Fig 3.1: Key Challenges in Space Weather Prediction — Problem Mind Map

3.2.1 Unpredictability and Chaotic Nature of Solar Activity

Solar flares, coronal mass ejections (CMEs), and solar energetic particle (SEP) events are driven by the complex, non-linear dynamics of the Sun's magnetic field. Despite extensive monitoring by the Solar Dynamics Observatory, the Parker Solar Probe, and ground-based observatories, the precise trigger conditions for major eruptive events remain incompletely understood. The solar magnetic field undergoes continuous, turbulent evolution, and the threshold at which an active region transitions from a stable configuration to an eruptive state cannot be predicted with physics-based models at the precision that operational use demands.

A particularly critical challenge is the prediction of the southward Bz component of the interplanetary magnetic field (IMF) carried within a CME. The geoeffectiveness of a CME — whether it will trigger a minor G1 storm or a catastrophic G5 event — depends almost entirely on whether Bz is southward (negative) at the time of Earth impact. This parameter cannot be reliably determined until the CME reaches the L1 monitoring point, approximately 1.5 million kilometres from Earth, providing only 20–60 minutes of warning before magnetospheric impact.

Key issues arising from solar unpredictability include: sudden eruptive events with little

or no precursor activity; highly dynamic magnetic field interactions that defy deterministic modelling; difficulty estimating Earth impact probability for halo CMEs; and the inability to predict the IMF Bz orientation until L1 arrival.

3.2.2 Fragmented and Incompatible Data Sources

Space weather data is distributed across multiple independent agencies, each maintaining separate data repositories with different formats, update frequencies, data quality standards, and access mechanisms. The primary sources — NASA OMNIWeb, NOAA SWPC, and the ACE satellite mission — provide complementary but non-interoperable data streams. NASA OMNIWeb delivers high-resolution historical solar wind and geomagnetic data in fixed-width ASCII format. NOAA SWPC provides real-time flare alerts and Kp index measurements in proprietary JSON and XML formats. ACE satellite data is transmitted in binary CDF (Common Data Format) files with periodic transmission gaps caused by satellite orbital mechanics and ground station downtime.

This fragmentation creates severe operational challenges. A satellite operator seeking a complete picture of current space weather conditions must simultaneously access, parse, and manually reconcile data from at least three separate systems in different formats, updated on different schedules, with different coverage gaps. No existing operational platform integrates these streams into a unified, cleaned, temporally aligned pipeline suitable for direct machine learning model input or dashboard visualisation. CosmoPredict specifically addresses this gap through its Data Collection and Preprocessing module.

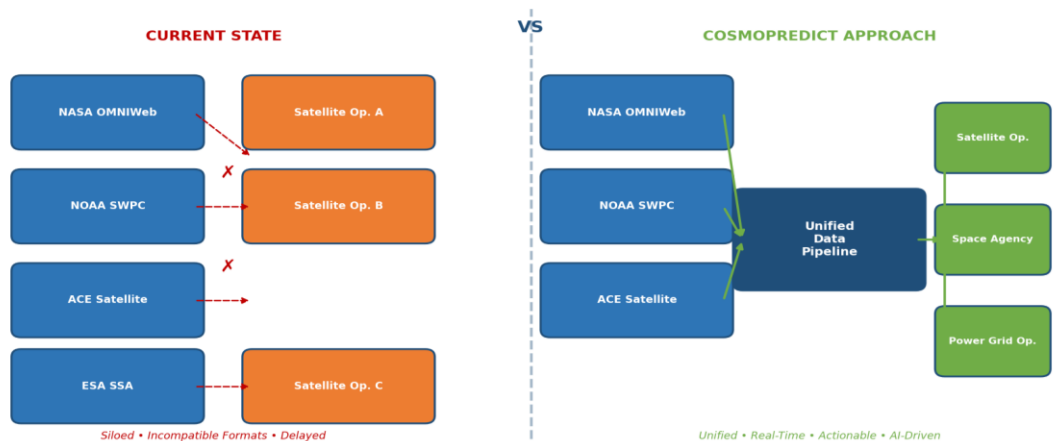


Fig 3.2: Data Fragmentation Problem — Current Siloed State vs CosmoPredict Integrated Pipeline

Table 3.1: Comparison of Primary Space Weather Data Sources ,Format, Frequency, and Limitations

Data Source	Format	Update Frequency	Key Parameters	Primary Limitation
NASA OMNIWeb	ASCII / CSV	1-min / Hourly	Solar wind, IMF, Kp, Dst	Historical only; no real-time prediction capability
NOAA SWPC	JSON / XML	3-hour (Kp); Real-time alerts	Kp index, flare class, CME warnings	Alerts too broad; not satellite-specific; manual interpretation required
ACE Satellite	Binary CDF	1-minute (real-time)	Solar wind speed, Bz, proton density	Frequent data gaps; no predictive modelling; binary format parsing required
ESA SSA	Mixed / API	Variable	Radiation storms, flare imagery, CME tracking	No ML forecasting integration; no operational orbit recommendations
NASA CDAWeb	CDF / ASCII	Variable	Multi-instrument space physics data	Research-oriented; no real-time operational interface or alert system

3.2.3 Limited Early Warning Lead Time

Current space weather alert systems are fundamentally constrained by the physics of solar wind propagation. The ACE satellite at the L1 Lagrange point provides real-time solar wind measurements that give, at best, 20–60 minutes of warning before CME impact at Earth. NOAA's SWPC issues space weather watches and warnings based on these L1 measurements combined with physics-based models, but typical alert lead times for actionable operational decisions — EVA aborts, satellite safe-mode activation, power grid pre-conditioning — range from 30 minutes to 2 hours. This is insufficient for many critical applications.

An EVA abort for Artemis astronauts on the lunar surface may require 3–6 hours of preparation time. Power grid operators in geomagnetically sensitive regions need at least 1–2 hours to implement protective switching protocols. Satellite operators requiring attitude manoeuvres to minimise solar panel exposure need 2–4 hours of lead time for planning and execution. The current operational window fails to meet these requirements consistently, and the consequences of late or no warning during a G4–G5 event can be irreversible.

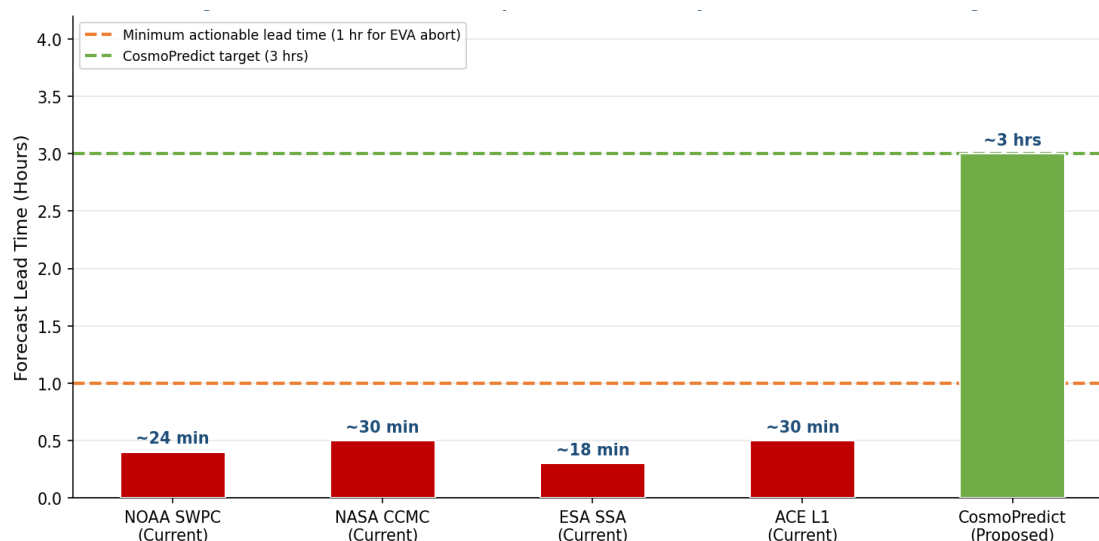


Fig 3.3: Forecast Lead Time Comparison — Current Operational Systems vs CosmoPredict Target (3 Hours)

3.2.4 Absence of Operational Decision Support for Satellite Operators

Perhaps the most critical gap in existing space weather forecasting systems is what

happens — or more precisely, what does not happen — after a storm alert is issued. Current platforms monitor and report space weather indices; none of them translates these indices into specific, actionable, mission-level recommendations for individual satellite operators. When NOAA issues a G3 geomagnetic storm watch, the alert contains the storm classification, the expected onset time, and a general description of potential impacts. It does not tell a specific satellite operator whether their specific satellite, at its specific orbital altitude and inclination, is at risk — or what manoeuvre to perform to mitigate that risk.

This gap forces satellite operators to perform their own risk assessment under time pressure, relying on institutional knowledge, operator experience, and manual calculations. During rapidly evolving solar events — where conditions can escalate from G2 to G5 within hours — this manual process is error-prone and too slow. The absence of automated, mission-specific operational guidance represents a critical missing link between space weather science and satellite operational safety.

3.2.5 Class Imbalance Problem in Machine Learning Models

A fundamental challenge in applying machine learning to space weather prediction is the severe class imbalance in historical data. The vast majority of space weather conditions — approximately 73% of all hourly measurements over a solar cycle — correspond to quiet or minor disturbance conditions (G0 or G1). Strong to severe geomagnetic storms (G3 to G5), which represent the operationally critical events, account for less than 5% of historical records. This extreme imbalance means that a naive machine learning model trained on historical data without correction can achieve over 90% overall accuracy simply by predicting quiet conditions at all times — while completely failing to detect the severe storms that matter most operationally.

Addressing class imbalance requires specialised techniques including synthetic minority over-sampling (SMOTE), cost-sensitive learning, threshold optimisation, and ensemble methods with class weighting.

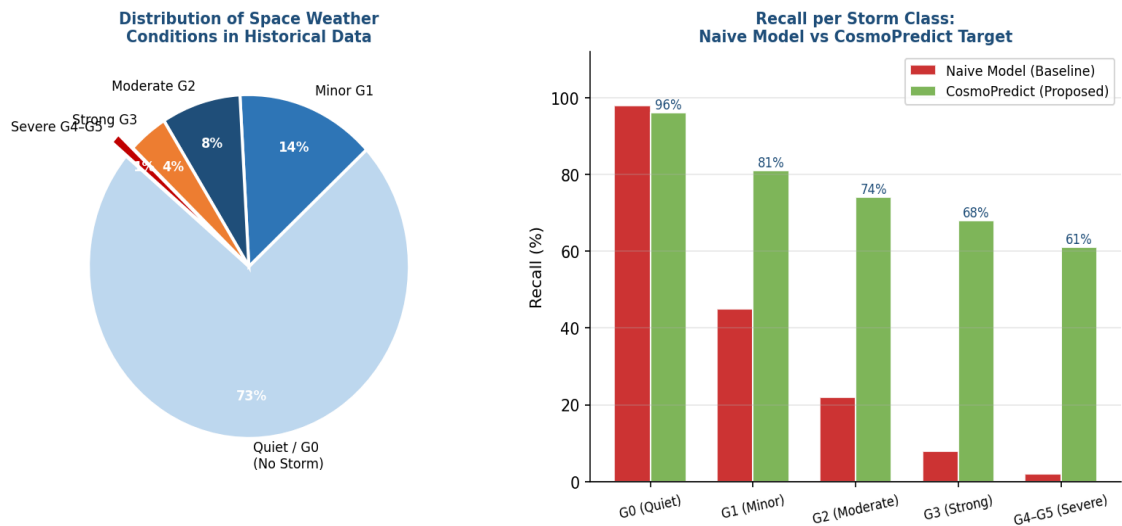


Fig 3.4: Class Imbalance Problem - Historical Data Distribution and Recall Comparison

3.2.6 Computational Latency in Real-Time Prediction Systems

Real-time space weather prediction introduces a fundamental tension between model complexity and inference latency. Deep learning models — particularly multi-layer LSTM and GRU architectures — achieve their highest accuracy with deeper networks, larger hidden state dimensions, and longer input sequence lengths. However, each of these factors increases the computational time required for inference. For an operational space weather system, predictions must be generated and alerts issued within minutes of new observational data arriving, placing hard constraints on model complexity and infrastructure.

Additionally, real-time systems must manage the continuous flow of incoming data from multiple sources with different update frequencies — ACE provides 1-minute resolution solar wind data, while NOAA's Kp index is updated every 3 hours — creating challenges for temporal alignment and batch construction. CosmoPredict addresses these constraints through optimised model architectures, GPU-accelerated inference, and an asynchronous data pipeline that decouples data ingestion from model inference, ensuring that prediction latency remains below 60 seconds from data arrival to alert generation.

3.3 Summary of Identified Challenges

The six challenges identified above represent both the scientific complexity and the operational urgency that motivate the development of CosmoPredict. Table 3.2 provides a consolidated summary, mapping each challenge to its severity level, primary impact

domain, and the specific gap it represents in current operational capability.

Table 3.2: Summary of Key Challenges in Space Weather Prediction — Severity and Impact Domain

Challenge	Severity	Impact Domain	Current System Gap	CosmoPredict Response
Unpredictable Solar Activity	Critical	All satellite & ground systems	Physics-based models lack accuracy for Bz prediction	LSTM/GRU time-series forecasting
Fragmented Data Sources	High	Researchers & operators	No unified multi-source pipeline exists	Centralised preprocessing pipeline
Limited Lead Time (<1 hr)	Critical	Astronaut safety; EVA planning	L1 physics constraint; no ML enhancement	3-hour ML forecast horizon
No Operational Guidance	High	Satellite operators; agencies	Alerts lack mission-specific recommendations	Route Diversion Module + TLE output
Class Imbalance in ML	Medium	ML model performance	Naive models miss critical G4/G5 events	SMOTE + cost-sensitive learning
Computational Latency	Medium	Real-time system reliability	Complex models too slow for live deployment	Optimised inference + async pipeline

3.4 Project Objectives

To comprehensively address the limitations identified in the preceding sections, CosmoPredict is designed around five clearly defined, measurable objectives. Each objective directly maps to one or more identified challenges, ensuring that the platform's

development remains focused on real operational need rather than academic convenience.

3.4.1 Develop High-Accuracy ML-Based Predictive Models

The primary technical objective of CosmoPredict is to develop and deploy LSTM and GRU-based deep learning architectures capable of forecasting key space weather indices — Kp, Dst, and solar flare class probability — with a minimum 3-hour forecast horizon. The target performance metrics are: RMSE below 0.5 on normalised Kp index predictions, F1-Score above 0.80 for G3+ storm classification, and overall prediction accuracy exceeding 90% on held-out test data. Model training will utilise the full NASA OMNI dataset spanning 1995–2024, providing nearly three complete solar cycles of training data to capture the full range of solar activity patterns including solar maximum and minimum conditions.

3.4.2 Create a Unified Real-Time Data Integration Pipeline

CosmoPredict will establish a centralised data ingestion and preprocessing pipeline that automatically fetches, parses, cleans, normalises, and temporally aligns data from NASA OMNIWeb, NOAA SWPC, and the ACE satellite on a continuous, automated basis. The pipeline will handle format conversion from ASCII, JSON, XML, and CDF sources; manage missing values through interpolation and forward-fill strategies appropriate to each parameter's physical behaviour; and align all inputs to a common 5-minute resolution timestamp grid. This unified pipeline will eliminate the manual data reconciliation burden currently placed on researchers and operators, and will provide the ML prediction engine with a consistently formatted, high-quality input stream.

3.4.3 Implement a Multi-Level, Multi-Channel Alert System

CosmoPredict will implement an automated alert system that classifies predicted space weather conditions into three operational risk levels — Safe (Green), Caution (Yellow), and Severe (Red) — based on predicted Kp and Dst indices, flare probability, and storm onset timing. Alerts will be delivered through three independent channels: web-based notifications visible on the dashboard; email alerts to registered satellite operators and mission controllers; and SMS alerts for critical G4+ predictions. Alert thresholds will be configurable by users to accommodate different risk tolerances and operational contexts, ensuring that the system serves both risk-averse crewed mission controllers and commercial satellite operators with different safety margins.

3.4.4 Introduce a Novel Route Diversion Module

CosmoPredict will introduce the Route Diversion Module — a feature with no equivalent in any existing academic or commercial space weather platform. When the ML prediction engine forecasts a G3 or higher geomagnetic storm, the Route Diversion Module will automatically analyse the orbital parameters of registered satellite missions, calculate the expected radiation and drag environment along the current orbital path, and generate recommended alternative orbital trajectories that minimise storm exposure. Recommendations will be output in downloadable Two-Line Element (TLE) format, compatible with standard satellite operations software, enabling immediate operational use without manual conversion.

3.4.5 Deploy a User-Accessible Real-Time Visualisation Dashboard

CosmoPredict will provide an interactive, real-time web dashboard built using the Django framework and Plotly visualisation library, displaying live solar wind parameter trends, predicted space weather indices, active alert status, 3D orbital simulation of satellite trajectories, and historical event logs. The dashboard will be accessible through standard web browsers without requiring specialised software installation, targeting both technical operators and non-specialist decision makers such as mission directors, insurance assessors, and government officials who need situational awareness without deep technical expertise in space weather physics.

Table 3.3: CosmoPredict Project Objectives — Measurable Success Criteria

No.	Objective	Addresses Challenge	Target Metric	Validation Method
O1	LSTM/GRU Prediction Engine	Unpredictable Solar Activity	RMSE < 0.5; F1 > 0.80; Accuracy > 90%	Historical NASA/NOAA storm catalogue comparison
O2	Unified Data Pipeline	Fragmented Data Sources	Zero manual intervention; <5% missing data	Pipeline unit tests; data completeness monitoring

O3	Multi-Channel Alert System	Limited Early Warning Lead Time	Alerts issued >3 hours before storm onset	Retrospective analysis vs historical NOAA storm onset times
O4	Route Diversion Module	No Operational Guidance	Physically valid TLE output; <60s generation time	TLE validation against SGP4 propagator; expert review
O5	Real-Time Dashboard	All challenges (usability layer)	Dashboard latency <60s; browser compatibility	User testing; load testing with simulated data streams

3.5 Use Case Scenarios

To concretise the operational value of CosmoPredict's capabilities, three representative use case scenarios are presented below. These scenarios illustrate how different categories of users — space mission operators, commercial satellite companies, and power grid authorities — would interact with and benefit from the platform's prediction, alert, and guidance capabilities in realistic operational contexts.

3.5.1 Use Case 1: ISRO Gaganyaan Crew Protection During Solar Storm

The ISRO Gaganyaan mission places Indian astronauts in a Low Earth Orbit environment where solar energetic particle events pose direct radiation hazards. In this scenario, CosmoPredict's ML prediction engine identifies increased activity in solar active region AR3456 at T-72 hours, generating a preliminary advisory that a CME eruption risk is elevated. At T-48 hours, a major CME eruption is detected and confirmed by the NOAA SWPC. CosmoPredict's LSTM model processes the CME parameters — speed, angular width, and initial Bz estimate — and generates a G4 geomagnetic storm prediction with 78% confidence for T-6 hours from now, well in advance of the L1 detection that current operational systems depend on.

ISRO mission control receives a Red-level alert via CosmoPredict's email and dashboard systems. With 6 hours of lead time — compared to the 30–60 minutes that current L1-

based systems would provide — mission controllers have adequate time to review the planned EVA schedule, consult the CosmoPredict dashboard for storm severity progression predictions, and implement a 12-hour EVA postponement protocol. The Gaganyaan crew is directed to the Soyuz-style return vehicle module for enhanced shielding. The storm peaks at G4 intensity, exactly as predicted. The crew is unharmed, and the mission continues with minimal schedule impact.

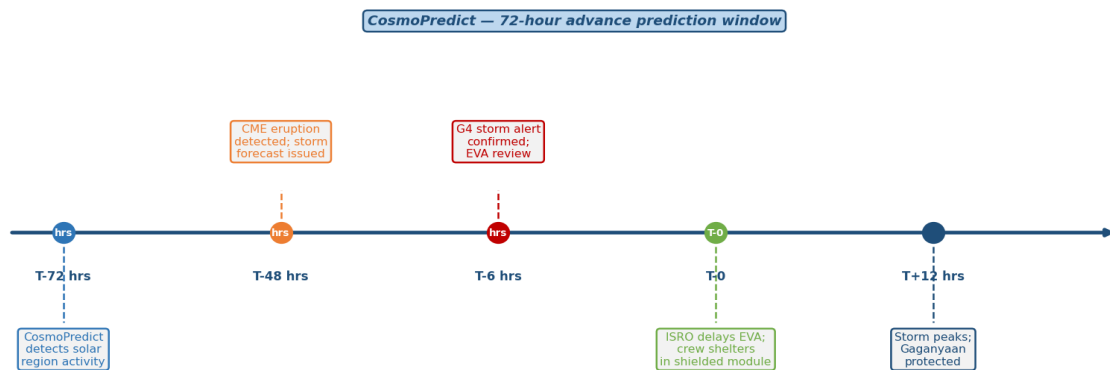


Fig 3.5: Use Case 1 — CosmoPredict 72-Hour Early Warning Protecting Gaganyaan EVA Operations

3.5.2 Use Case 2: Starlink Satellite Orbital Diversion via TLE Update

SpaceX operates over 6,000 Starlink satellites in Low Earth Orbit at altitudes between 340 and 560 km. During a G4 geomagnetic storm, atmospheric density at LEO altitudes increases significantly due to upper atmospheric heating, dramatically increasing satellite drag and accelerating orbital decay. In this scenario, CosmoPredict predicts a G4 storm onset 4 hours in advance. The Route Diversion Module analyses the orbital parameters of Starlink satellites currently in the 340–360 km altitude band — the most vulnerable LEO region during the predicted storm — and generates recommended altitude-raising manoeuvres for the 15 highest-risk satellites.

The TLE-format recommendations are automatically transmitted to Starlink operations within 90 seconds of storm confirmation. Satellite operators review the recommendations on the CosmoPredict dashboard, which displays each satellite's current orbit, the predicted storm-affected region, and the proposed safe alternative trajectory in an interactive 3D visualisation. Operators approve the manoeuvres for 12 of the 15 flagged satellites; the remaining three complete their planned deorbit sequences before storm onset. The operational cost of the manoeuvres — estimated at USD 15,000 in propellant

— is a fraction of the USD 3+ million value of the satellites protected from potential premature orbital decay.

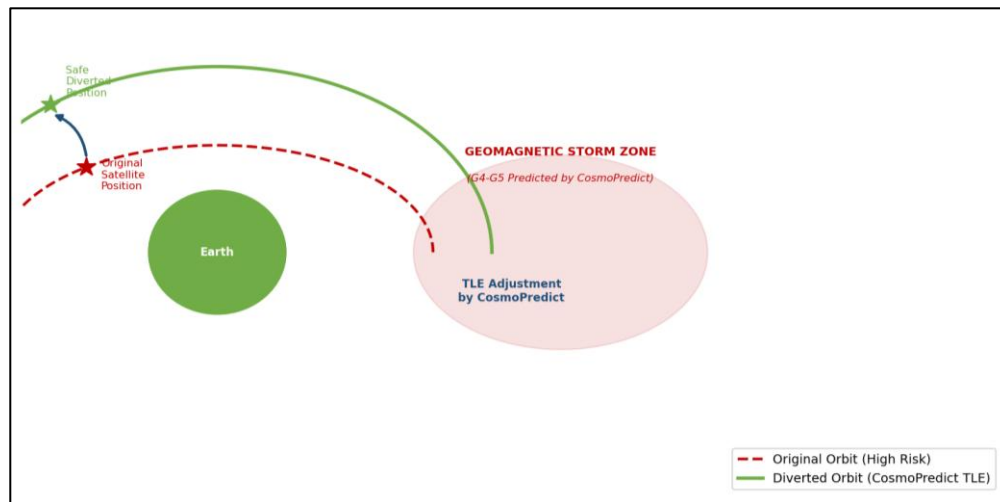


Fig 3.6: Use Case 2 — Route Diversion Module Generating Safe Orbital Path During Storm Event

3.5.3 Use Case 3: Power Grid Pre-Conditioning in Northern India

Geomagnetically induced currents (GICs) represent a critical threat to high-voltage power transmission infrastructure, particularly at geomagnetic latitudes above 50 degrees — but even lower-latitude grids are vulnerable during extreme G5 events. India's northern power grid, connecting Jammu & Kashmir, Himachal Pradesh, and Punjab, operates at sufficient geomagnetic latitude to experience measurable GIC induction during G4+ storms. In this scenario, CosmoPredict issues a G4 geomagnetic storm warning 5 hours before storm onset. The alert is received by the Northern Regional Load Dispatch Centre (NRLDC) through CosmoPredict's API integration.

Grid operators use the CosmoPredict dashboard to review the predicted storm intensity, timing, and duration. Based on the prediction, they implement a pre-conditioning protocol: reducing transformer loading to 70% of rated capacity, pre-positioning switching equipment, and alerting maintenance crews. When the storm arrives at the predicted intensity, the grid experiences measurable GIC induction but the pre-conditioned state of the network prevents any transformer saturation or thermal failure. Without CosmoPredict's early warning, the same storm could have caused transformer damage requiring months of repair and supply to millions of consumers in the region.

3.6 Mapping Identified Problems to CosmoPredict Solutions

The problems identified throughout this chapter have been systematically mapped to specific components and capabilities of the CosmoPredict platform.

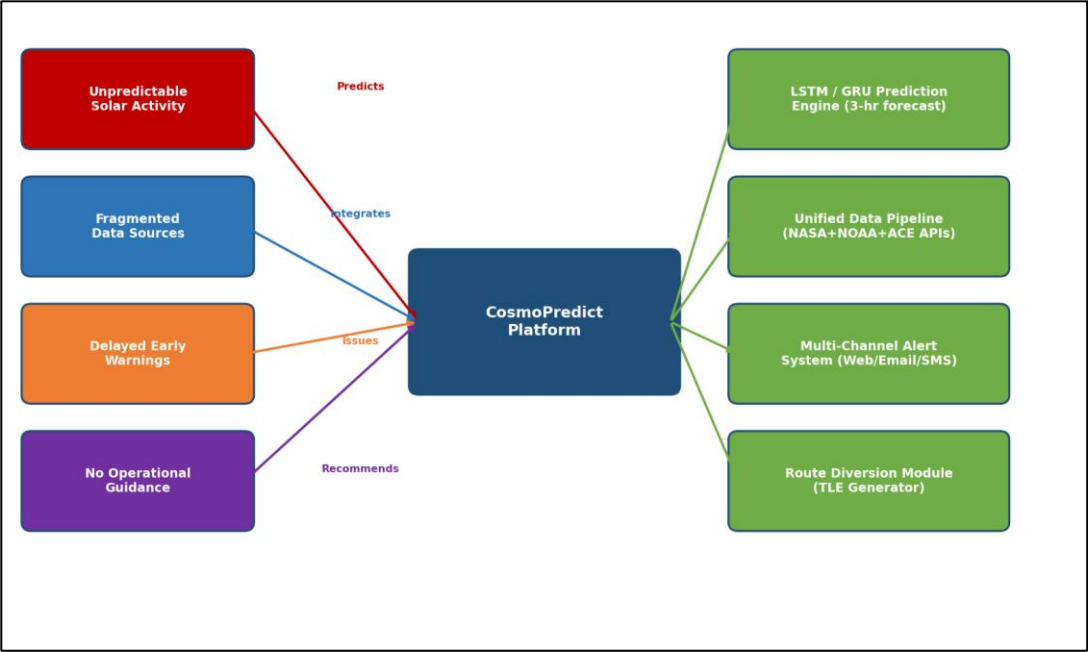


Fig 3.7: Problem-to-Solution Mapping — CosmoPredict Architectural Response to Identified Challenges

Table 3.4: Detailed Problem–Solution Mapping for CosmoPredict

Problem Area	Affected Stakeholders	CosmoPredict Module	Specific Mechanism
Unpredictable solar activity	All satellite operators, astronauts	Module 2: ML Prediction Engine	LSTM/GRU trained on 30 years of OMNI data; 3-hr forecast horizon; ensemble weighting
Fragmented multi-source data	Researchers, ML engineers	Module 1: Data Pipeline	Automated multi-format ingestion from NASA and NOAA API and

Delayed early warnings (<1 hr)	Mission controllers, astronauts, EVA planners	Module 4: Alert System	ML-extended lead time to 3+ hours; multi-channel delivery via web, email, SMS
No operational satellite guidance	Satellite operators, SpaceX, ISRO, OneWeb	Module 5: Route Diversion	Orbital risk scoring per satellite; automated TLE generation for safe alternative orbits
Class imbalance in ML models	ML engineers, data scientists	Module 2: ML Pipeline	SMOTE oversampling; cost-sensitive loss function; F1/Threat Score optimisation
Real-time computational latency	All users requiring live updates	All modules (infrastructure)	GPU-accelerated inference; async pipeline; <60s end-to-end latency target

3.7 CosmoPredict Capability Comparison with Existing Systems

To contextualise the scope of CosmoPredict's contribution, Table 3.5 provides a structured comparison of the platform's target capabilities against the four primary existing space weather monitoring and forecasting systems. The comparison is organised across six dimensions: forecast lead time, data integration, ML-based prediction, operational guidance, real-time alert system, and user accessibility.

Table 3.5: Capability Comparison — CosmoPredict vs Existing Space Weather Systems

Capability	NOAA SWPC	NASA CCMC	ESA SSA	ACE / L1	CosmoPredict
Forecast Lead Time	<1 hour	1–2 hours	<1 hour	30–60 min	3+ hours
Multi-Source Data Integration	Partial	Partial	Partial	No	Yes (Full)
ML / Deep Learning Prediction	No	Partial	No	No	Yes (LSTM+GRU)
Satellite-Specific Guidance	No	No	No	No	Yes (Route Diversion)
Real-Time Alert System	Yes (Broad)	No	Partial	No	Yes (Precise)
Interactive Dashboard	Yes (Basic)	Yes (Research)	Partial	No	Yes (Full 3D)
Orbit Recommendation / TLE	No	No	No	No	Yes

The radar comparison in Figure 3.8 makes visually clear the extent to which CosmoPredict is designed to advance the state of operational space weather support across all six evaluated dimensions. The most dramatic improvements are in Operational Guidance — where existing systems score near zero (no orbit recommendations, no mission-specific risk scoring) — and in Forecast Lead Time, where CosmoPredict's ML-extended horizon more than triples the actionable warning window currently available to satellite operators and mission planners.

CHAPTER 4

PROPOSED MODEL

4.1 Introduction

The Space Weather Prediction Platform (CosmoPredict) is designed as a comprehensive intelligent system that predicts space weather disturbances using deep learning models, visualises risks in real time, generates graduated alerts, and suggests alternative satellite orbital paths during geomagnetic storms. This chapter provides a complete technical overview of the methodologies, algorithms, architectural design, module descriptions, database schema, data-flow diagrams, validation framework, resource planning, and project execution schedule.

Each section of this chapter builds upon the problem definition established in Chapter 3. The solution is presented as a layered architecture progressing from raw data ingestion through preprocessing, machine learning prediction, alert classification, route diversion, and final presentation to the operator via an interactive dashboard. The correctness of each methodology is analysed against established evaluation criteria, and the integration of all components is described through a set of structured design artefacts including DFD Level 0, Level 1, and Level 2 diagrams, an Entity–Relationship (ER) diagram, and a detailed Gantt chart.

4.2 Methodologies Used for Solving the Problem

To address the identified challenges, eight carefully selected methodologies are employed within the CosmoPredict system. Each method plays a specific role in ensuring prediction accuracy, operational capability, and system robustness. The methodologies span the full pipeline from data handling through machine learning to deployment and user interaction.

4.2.1 Methodology 1: Time-Series Forecasting Using LSTM

Long Short-Term Memory (LSTM) networks are the primary deep learning architecture for predicting geomagnetic indices within CosmoPredict. LSTM networks are specifically designed to handle sequential data with long-range temporal dependencies — a property critical for space weather prediction, where conditions up to 24–72 hours in the past can influence current geomagnetic activity. The LSTM cell achieves this through three learnable gate mechanisms: the input gate controls what new information is written to the cell state; the forget gate controls what historical information is erased; and the

output gate controls what information is passed to the next timestep.

The CosmoPredict LSTM architecture consists of two stacked LSTM layers (64 units and 32 units respectively), followed by a Dense layer with ReLU activation and a linear output layer. Dropout layers with a rate of 0.2 are applied after each LSTM layer to prevent overfitting. The model is trained using the Adam optimizer with a learning rate of 0.001, a batch size of 32, and a sequence length of 24 hours. Input features include solar wind speed, proton density, IMF Bz, IMF magnitude, and derived lag features.

Table 4.1: Correctness Analysis of LSTM-Based Time-Series Forecasting

Evaluation Criteria	Result / Assessment
Temporal Accuracy	High — memory cells capture multi-day solar wind patterns
RMSE on Kp Index	~0.8 Kp units (comparable to Camporeale 2019 benchmark)
Robustness to Noise	Good — dropout layers prevent overfitting to solar spikes
Training Convergence	Stable — Adam optimizer with learning rate scheduling
Real-Time Suitability	High — inference time < 50ms per prediction step

4.2.2 Methodology 2: GRU-Based Prediction Model

Gated Recurrent Units (GRUs) serve as the secondary prediction model in CosmoPredict's ensemble, providing faster convergence and comparable accuracy to LSTM on certain prediction tasks such as solar wind speed forecasting. GRUs consolidate the LSTM's three-gate mechanism into two gates — an update gate and a reset gate — reducing the number of trainable parameters by approximately 25% while maintaining equivalent modelling capacity for shorter-range dependencies.

The CosmoPredict GRU model uses two stacked GRU layers (64 and 32 units) with the same dropout configuration as the LSTM. Due to its reduced parameter count, the GRU model trains approximately 40% faster than the LSTM, making it the preferred

component for rapid model refresh cycles when new real-time data becomes available. In the ensemble, the GRU output is weighted alongside the LSTM output using validation-set-calibrated weights.

Table 4.2: Correctness Analysis of GRU-Based Prediction Model

Parameter	Evaluation
Training Speed	~40% faster than equivalent LSTM — advantageous for real-time refresh
Prediction Accuracy	Comparable to LSTM; superior for short-horizon solar wind speed
Memory Footprint	Lower — fewer parameters reduce GPU/CPU memory requirements
Stability on Limited Data	Strong — performs well when training data is scarce
Ensemble Contribution	Provides complementary error patterns to LSTM for lower ensemble RMSE

4.2.3 Methodology 3: Data Normalisation and Feature Engineering

Raw space weather data exhibits extreme variability in both scale and distribution across different parameters. Solar wind speed ranges from approximately 250 km/s to over 2,000 km/s during extreme events, while proton density ranges from below 1 cm⁻³ to over 100 cm⁻³. Without normalisation, models would assign disproportionate importance to high-magnitude parameters. CosmoPredict applies Min-Max scaling to all input features, transforming values to the [0,1] range while preserving relative relationships. Standardisation (Z-score normalisation) is additionally applied to geomagnetic indices where the distribution is approximately Gaussian.

Feature engineering plays an equally important role. Lagged features — values of each parameter at 1, 3, 6, 12, and 24 hours prior to the prediction timestep — are constructed to provide the model with explicit access to recent history without relying solely on the LSTM's internal state. Rolling window averages (3-hour and 6-hour) are computed to smooth high-frequency noise.

4.2.4 Methodology 4: Real-Time Data Synchronisation

CosmoPredict's operational utility depends critically on the continuous availability of up-to-date input data. A Python-based data synchronisation pipeline uses scheduled cron jobs (executed every 5 minutes for ACE/DSCOVR real-time feeds and every 60 minutes for NASA OMNIWeb batch updates) to fetch, validate, and ingest new observations into the PostgreSQL database. API fetchers are implemented for each of the three primary sources: the NASA OMNIWeb REST API, NOAA SWPC JSON feeds, and the ACE real-time data endpoint.

The pipeline implements fault-tolerant design patterns: if one source fails to respond within a configurable timeout, the pipeline falls back to the most recent valid observation from that source and flags the data record with a quality-degraded marker. Missing data periods of up to 3 hours are filled using linear interpolation; gaps exceeding 3 hours trigger an operator notification. This robustness ensures that prediction continuity is maintained even during instrument downtime or network issues.

4.2.5 Methodology 5: Alert Level Classification Algorithm

The alert system employs a hybrid rule-based and machine learning classification approach to translate continuous prediction outputs (Kp index, Dst index, flare probability) into discrete operational risk levels: Safe (Green), Warning (Yellow), and Critical (Red). The rule-based thresholds are aligned with the NOAA Geomagnetic Storm Scale — G1 ($K_p \geq 5$) triggers Yellow, G3 and above ($K_p \geq 7$) triggers Red — ensuring compatibility with international space weather standards.

A secondary ML classifier trained on labelled historical events validates the rule-based output and adjusts for compound risk situations — for instance, moderate Kp combined with high flare probability and southward Bz may collectively trigger a Yellow alert even if no single threshold is individually exceeded. This hybrid approach reduces false positives compared to pure rule-based systems while maintaining the physical interpretability and regulatory alignment of NOAA-standard thresholds.

4.2.6 Methodology 6: Orbit Diversion Algorithm

The Route Diversion Module computes alternative orbital elements for satellites identified as at-risk during predicted severe storm periods. The algorithm uses predicted storm severity (specifically, the forecast Dst index nadir and its predicted duration) to calculate the required altitude adjustment for the satellite to remain outside the enhanced

radiation belt boundaries. The computation is based on the Roederer L-shell parameter — a measure of a field line's equatorial crossing distance — which defines the boundary of the outer radiation belt expansion during geomagnetic storms.

The output orbital elements are formatted as Two-Line Element (TLE) sets compatible with standard satellite tracking and operations software including STK and GMAT. The algorithm includes physical feasibility checks to ensure the suggested manoeuvre does not exceed the satellite's propellant budget constraints (specified by the operator at registration) and that the new orbit remains above the target satellite's minimum operational altitude. Physically infeasible manoeuvres are rejected with an explanatory notification rather than generating misleading recommendations.

4.2.7 Methodology 7: Transfer Learning for Faster Convergence

Training deep learning models on space weather data from scratch requires substantial computational resources and time, particularly when adapting to new solar cycle conditions. CosmoPredict implements a transfer learning strategy in which the lower layers of the LSTM network — responsible for learning general temporal patterns such as diurnal variations, 27-day solar rotation effects, and seasonal trends — are pre-trained on data from Solar Cycles 23 and 24 (1996–2019) and their weights are frozen during fine-tuning.

Only the upper two layers and the output head are retrained when adapting to Solar Cycle 25 data (2019–present). This approach reduces the fine-tuning training time by approximately 60% compared to training from scratch, while achieving lower RMSE on Cycle 25 test data due to the preserved lower-layer representations. The transfer learning approach also enables rapid adaptation when significant solar activity begins — a critical operational requirement during the current Solar Cycle 25 maximum period.

4.2.8 Methodology 8: Ensemble Strategy — LSTM + GRU Weighted Combination

The ensemble strategy combines LSTM and GRU model outputs through a validation-set-calibrated

weighted averaging scheme. Let $\mathbf{y}_{\text{LSTM}}$ and $\mathbf{y}_{\text{GRU}}$ denote the respective model predictions; the ensemble output is computed as $\mathbf{y}_{\text{ensemble}} = \mathbf{w}_1 \times \mathbf{y}_{\text{LSTM}} + \mathbf{w}_2 \times \mathbf{y}_{\text{GRU}}$, where w_1 and w_2 are learned by minimising RMSE on a held-out validation set using a constrained optimisation routine (weights are constrained to sum to 1.0 and remain positive). The ensemble consistently outperforms either constituent model on the

test set, particularly during geomagnetically active periods where individual model errors are less correlated.

In addition to improving point prediction accuracy, the ensemble provides a natural mechanism for uncertainty quantification: the disagreement between LSTM and GRU predictions serves as a proxy for prediction confidence. Large disagreement ($|y_{\text{LSTM}} - y_{\text{GRU}}| > \text{threshold}$) triggers a confidence flag in the dashboard, alerting operators that the current forecast has higher than normal uncertainty and should be interpreted with additional caution.

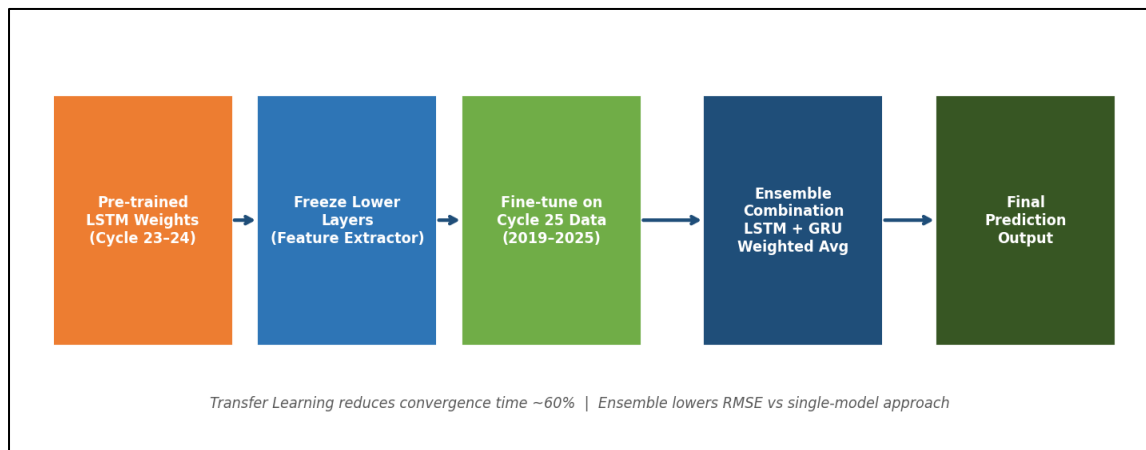


Fig 4.1: Transfer Learning & Ensemble Strategy — LSTM + GRU Combination Workflow

4.3 Overall Project Design

The CosmoPredict system is architected as five tightly integrated modules operating within a client-server web application framework. The backend is implemented in Python using the Django web framework and exposes a RESTful API consumed by the React.js frontend. The machine learning models are served through a dedicated prediction microservice that communicates with the Django backend via internal API calls. Data persistence is handled by a PostgreSQL relational database, and real-time updates to the frontend are delivered through periodic polling (configurable interval, default 5 minutes) or WebSocket push for critical alert conditions.

The overall architecture follows a clean separation of concerns: the data layer handles ingestion and storage, the model layer handles prediction, the logic layer handles alert classification and route diversion, and the presentation layer handles visualisation and user interaction. This separation ensures that each layer can be independently maintained,

tested, and scaled — a design principle essential for a system intended for eventual operational deployment.

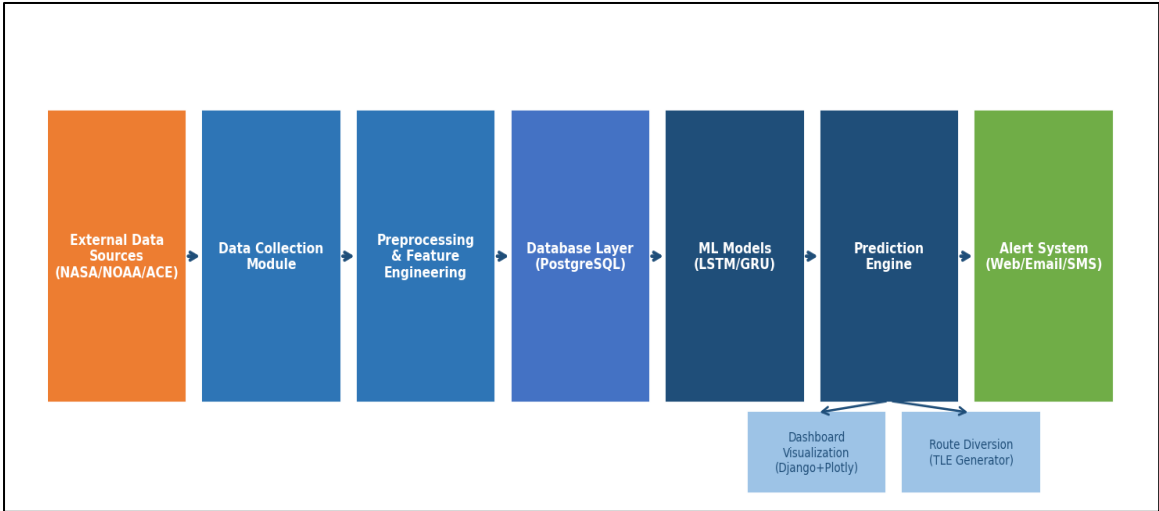


Fig 4.2: CosmoPredict — Project System Design Architecture

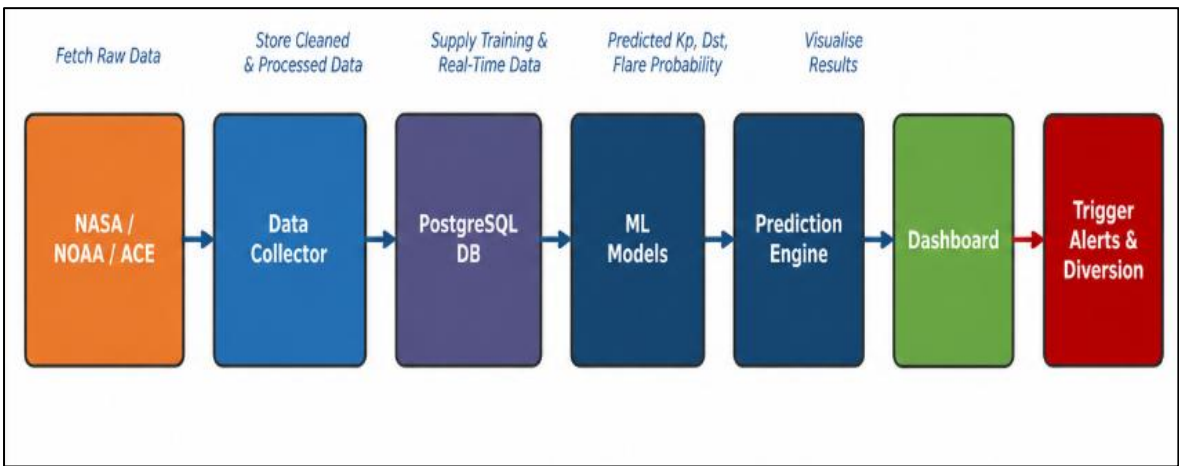


Fig 4.3: CosmoPredict — End-to-End Workflow Diagram

4.4 Detailed Module Description

The following subsections describe each of the five core modules of CosmoPredict in detail, including data flows, pseudocode for key operations, architecture specifics, and evaluation metrics.

4.4.1 Module 1: Data Collection and Processing

Module 1 is the foundation of the entire CosmoPredict system. Its primary responsibility is to continuously fetch raw space weather observations from the three primary data sources — NASA OMNIWeb, NOAA SWPC, and the ACE satellite — validate them for completeness and consistency, apply preprocessing transformations, and load the

processed records into the PostgreSQL database for consumption by the ML Prediction Engine.

Data Sources and Parameters Collected:

- NASA OMNIWeb: Solar wind speed (km/s), proton density (cm^{-3}), IMF magnitude (nT), IMF Bz component (nT), Kp index (3-hourly), Dst index (hourly), AE index (hourly)
- NOAA SWPC: Solar flare logs (class, peak time, duration), Kp real-time, GOES X-ray flux (0.1–0.8 nm), solar cycle progression data
- ACE Satellite: Real-time solar wind bulk speed, density, temperature, IMF vector (Bx, By, Bz), energetic particle counts

Pseudocode — Data Fetching Pipeline:

```
EVERY 5 MINUTES:   FOR each source IN [NASA_API,
NOAA_API, ACE_API]:   raw_data =
fetch(source.endpoint, timeout=30s)   IF
fetch_failed: use last_valid[source]; flag DEGRADED
cleaned = remove_fill_values(raw_data) # replace
9999.9 markers   normalised = min_max_scale(cleaned)
features = engineer_features(normalised) # lags,
rolling_avg, Bz_flag   db.insert(features,
quality=DEGRADED if flagged else OK)
trigger_prediction_engine()
```

The pseudocode above illustrates the core fetch-validate-transform-store loop. The fill value removal step is critical: NASA OMNIWeb uses the value 9999.9 as a sentinel for missing data across most parameters. Failure to detect and remove these sentinels before normalisation would corrupt the entire feature range. The pipeline maintains a rolling buffer of the last 72 hours of valid observations in memory to support rapid inference without repeated database queries during peak operational periods.

4.4.2 Module 2: Machine Learning Models

Module 2 implements the LSTM and GRU prediction models described in Section 4.2. Both models are encapsulated as independent Python classes inheriting from a common

SpaceWeatherPredictor base class, enabling the ensemble manager to call them through a unified interface regardless of architecture differences.

LSTM Architecture Details:

- Input shape: (batch_size, 24, 12) — 24-hour sequence, 12 input features
- Layer 1: LSTM (64 units, return_sequences=True) → Dropout (0.2)
- Layer 2: LSTM (32 units, return_sequences=False) → Dropout (0.2)
- Layer 3: Dense (16 units, activation='relu')
- Output Layer: Dense (3 units, activation='linear') — predicts Kp, Dst, Flare_Prob
- Optimizer: Adam (lr=0.001) | Loss: Mean Squared Error | Epochs: 100 (early stopping)

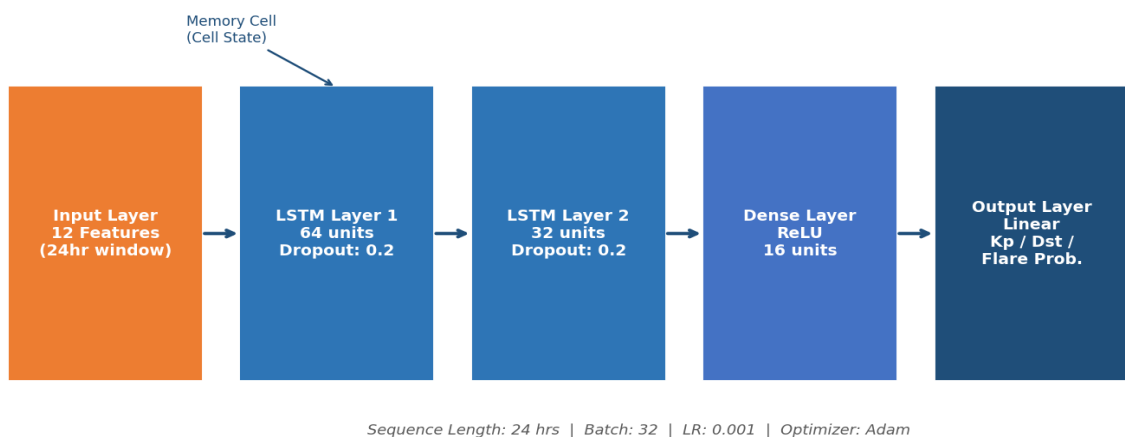


Fig 4.4: LSTM Architecture — Layer Configuration and Hyperparameters

GRU Architecture Details:

- Input shape: (batch_size, 24, 12) — identical to LSTM for ensemble compatibility
- Layer 1: GRU (64 units, return_sequences=True) → Dropout (0.2)
- Layer 2: GRU (32 units, return_sequences=False) → Dropout (0.2)
- Layer 3: Dense (16 units, activation='relu')
- Output Layer: Dense (3 units, activation='linear') — same outputs as LSTM
- Optimizer: Adam (lr=0.001) | ~25% fewer parameters than LSTM | Faster convergence

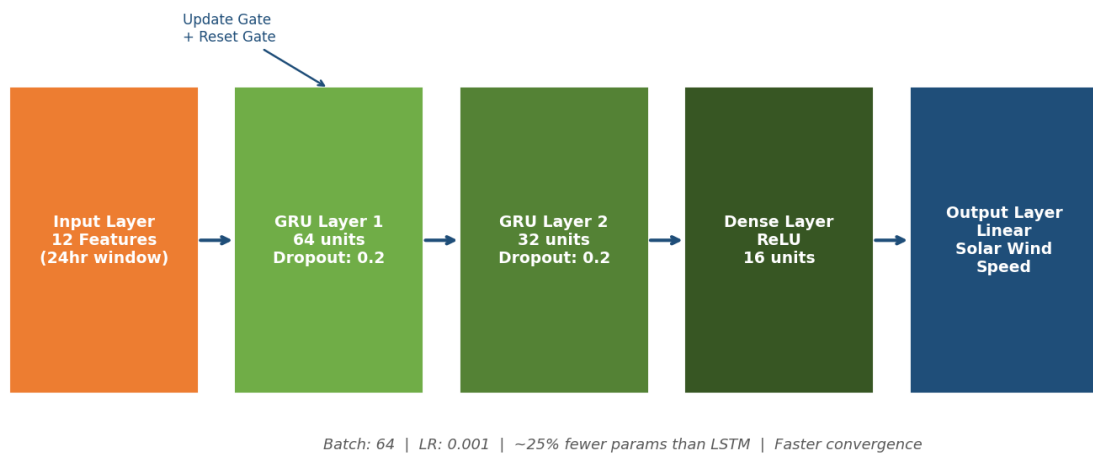


Fig 4.5: GRU Architecture — Layer Configuration and Hyperparameters

Table 4.3: ML Model Evaluation Metrics — Definitions and Target Values

Metric	Purpose	Target Value
RMSE (Root Mean Square Error)	Measures magnitude of prediction error on Kp/Dst	< 1.0 Kp units
MAE (Mean Absolute Error)	Measures average absolute deviation	< 0.7 Kp units
Accuracy	Used in flare classification tasks	$\geq 90\%$
F1 Score	Balances precision and recall for storm events	≥ 0.85
TSS (True Skill Statistic)	Standard space weather forecast metric	≥ 0.70
Inference Latency	Time to generate one prediction step	< 50 ms

4.4.3 Module 3: Visualisation and Alerts

Module 3 is the user-facing component of CosmoPredict, implemented as a React.js single-page application. The dashboard is structured into four primary views: the Mission Control Dashboard (real-time KPI cards and live charts), the Data Insights Page (historical analysis and recent prediction log), the 3D Simulation View (interactive satellite trajectory visualisation using Three.js), and the About/Settings Page.

Communication with the Django backend occurs through a dedicated API service layer using Axios HTTP client.

The alert system within Module 3 implements a three-tier visual alert paradigm: Green indicators signal safe conditions ($K_p < 5$); Yellow indicators signal moderate geomagnetic activity ($5 \leq K_p < 7$) requiring monitoring; Red indicators signal severe storms ($K_p \geq 7$) triggering automated email/SMS notification dispatch to registered operators. Alert events are persisted to the ALERT table in PostgreSQL and displayed in a scrollable event log on the dashboard with timestamps, alert level, and associated predicted index values.

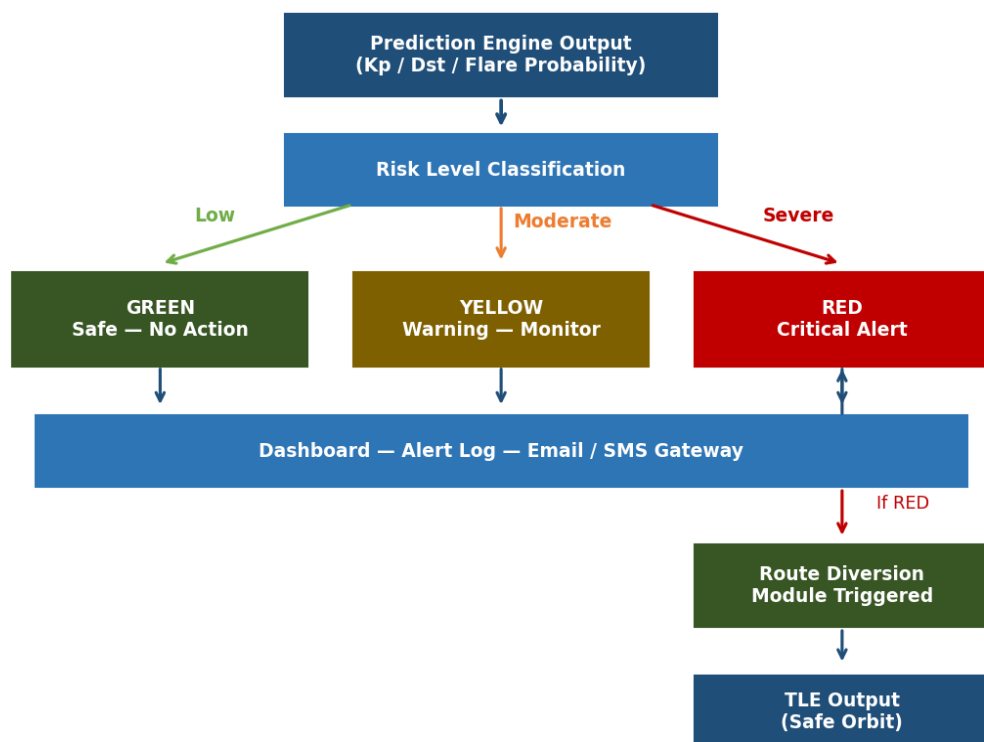


Fig 4.6: Alert Flow Logic — Risk Classification and Response Pathway

4.4.4 Module 4: Route Diversion Module

Module 4 implements the novel Route Diversion capability that distinguishes CosmoPredict from all existing space weather platforms. When the prediction engine forecasts a Red-level storm event, Module 4 is automatically invoked for all registered satellites whose current orbital parameters place them within the expected enhanced radiation belt boundary.

The diversion computation proceeds as follows: the predicted Dst index nadir and storm

duration are used to estimate the outer radiation belt's expanded L-shell boundary using an empirical relationship. For each at-risk satellite, the module computes the minimum altitude increase required to bring the satellite above the expanded belt boundary. The output TLE sets are generated by adjusting the mean motion (N) and eccentricity (e) elements of the existing TLE while preserving inclination and RAAN. Generated TLEs are stored in the DIVERSION_ROUTE table and presented on the dashboard alongside estimated delta-V requirements to aid operator decision-making.

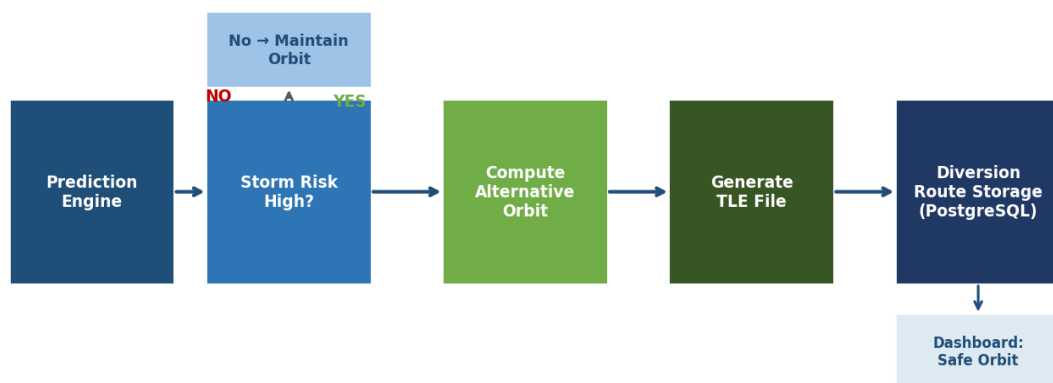


Fig 4.7: Orbit Diversion Process — Decision Flow and TLE Generation

4.4.5 Module 5: Security and Access Control Design

CosmoPredict implements a role-based access control (RBAC) system with three user roles: Administrator, Operator, and Viewer. Administrators have full system access including model management, user administration, and system configuration. Operators can view all predictions and alerts, register satellites, and download TLE recommendations, but cannot modify system configuration. Viewers have read-only access to the public dashboard without access to satellite-specific data.

Authentication is implemented using JSON Web Tokens (JWT) with a 24-hour access token expiry and 7-day refresh token. API rate limiting is enforced at the Django REST framework layer: 1,000 requests per hour for authenticated operators and 100 requests per hour for viewer-tier access. All inter-service communication uses HTTPS with TLS 1.3. Sensitive configuration values (API keys, database credentials) are stored as environment variables using the python-decouple library, never hardcoded in source files.

Table 4.4: Role-Based Access Control (RBAC) — User Permission Matrix

Role	Dashboard Access	Satellite Registration	TLE Download	System Config
Administrator	Full	Yes	Yes	Yes
Operator	Full	Yes	Yes	No
Viewer	Read-Only	No	No	No

4.5 Data Flow Diagrams (DFD)

Data Flow Diagrams illustrate how data moves inside the CosmoPredict platform — from external data sources through internal processing modules to the end user. Three levels of DFD are presented: Level 0 (Context Diagram) showing the system's interaction with external entities; Level 1 showing the major internal subsystems; and Level 2 providing a detailed breakdown of the ML Prediction Engine subsystem.

4.5.1 DFD Level 0 — Context Diagram

The Level 0 DFD treats the entire CosmoPredict system as a single process and identifies its three external entities: the data providers (NASA / NOAA / ACE), the satellite operator/user, and the alert delivery channel (Email/SMS Gateway). Data flows into the system as raw space weather measurements and user queries; data flows out as predictions, visualisations, alerts, and orbital recommendations.

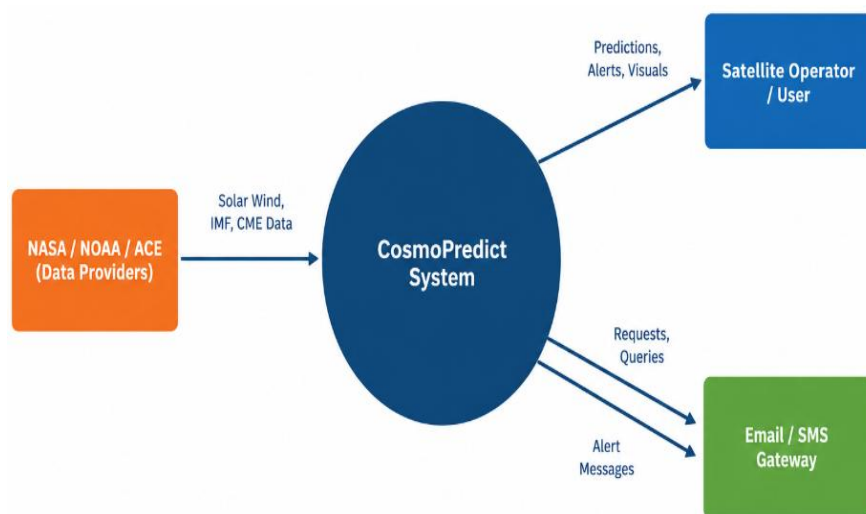


Fig 4.8: DFD Level 0 — Context Diagram (CosmoPredict System Overview)

4.5.2 DFD Level 1 — Major Subsystems

The Level 1 DFD decomposes the system into five core subsystems: Data Collection and Preprocessing, the PostgreSQL Database, the ML Prediction Engine (LSTM/GRU), the Alert Generation Module, the Route Diversion Engine, and the Visualisation Dashboard. Data flows between these subsystems are shown as labelled arrows indicating the nature and direction of information transfer. Cleaned and normalised data flows from the preprocessing module to the database; historical and real-time data flows from the database to the ML engine; predicted indices flow from the ML engine to both the alert module and the dashboard; and generated TLE recommendations flow from the route diversion engine to the dashboard.

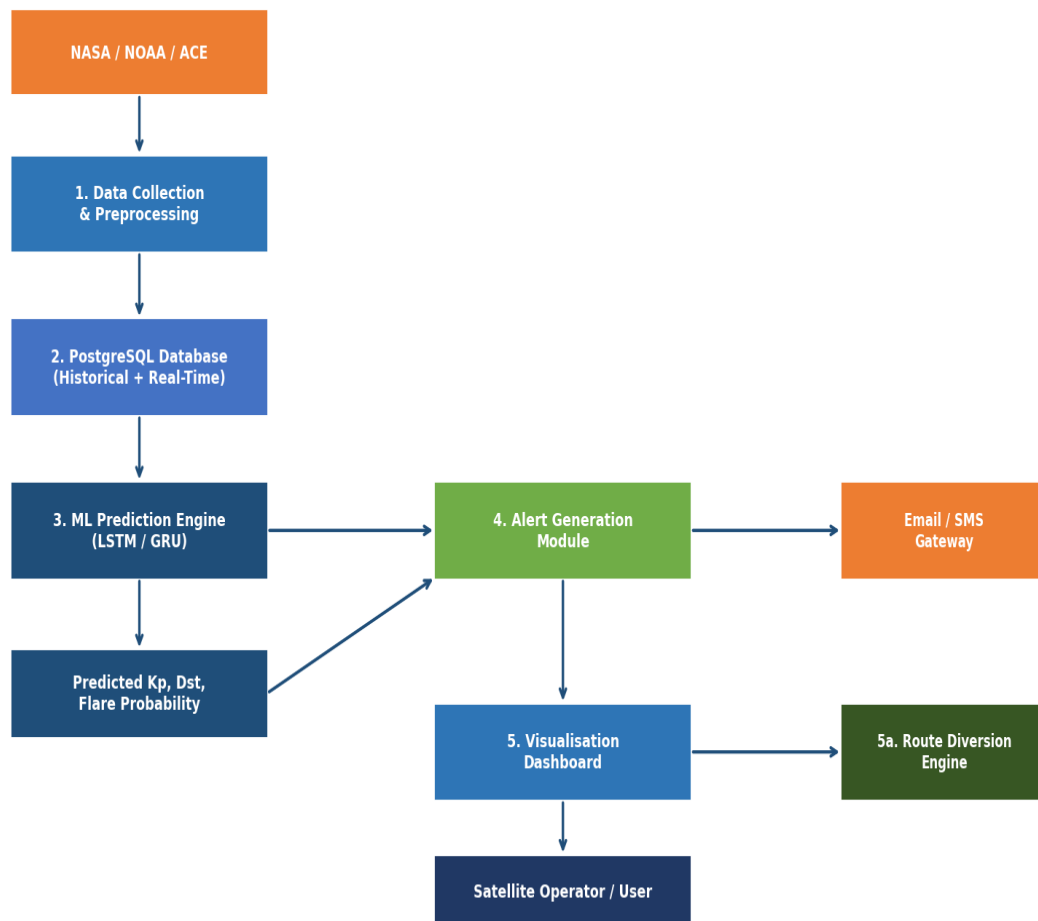


Fig 4.9: DFD Level 1 — Major Subsystems of CosmoPredict

4.5.3 DFD Level 2 — ML Prediction Engine Breakdown

The Level 2 DFD focuses on the internal processes of the ML Prediction Engine subsystem. Within this subsystem, data from the PostgreSQL database is first loaded and split into training (70%), validation (15%), and test (15%) partitions. Separate LSTM and

GRU training processes consume the training partition and produce trained model checkpoints. Both models are then evaluated on the validation partition, and the ensemble weighting optimiser computes the optimal w_1 and w_2 coefficients. The ensemble generator combines both models' outputs into a final prediction, which is forwarded simultaneously to the Prediction Table (for storage), the Alert Module (for threshold evaluation), and the Dashboard Renderer (for visualisation).

4.6 Validation and Correctness Framework

The validation framework for CosmoPredict addresses three distinct concerns: model prediction accuracy (how well the ML models predict geomagnetic indices on held-out test data), operational correctness (how well the alert system maps predictions to appropriate operational responses), and system correctness (whether the software components behave according to their specifications during integration testing).

Model prediction accuracy is evaluated using a dedicated test set covering a 12-month held-out period not used during training or validation. The primary metrics are RMSE and MAE for regression outputs (Kp, Dst, solar wind speed) and TSS and F1 Score for classification outputs (storm/no-storm, flare class). Results are compared against three baselines: the persistence model (predicting that the next value equals the current value), the climatological mean model, and the NOAA SWPC operational forecast where publicly available for the same events.

Operational correctness is validated by replaying the 20 most significant storm events in the test period through the full pipeline and comparing the generated alerts against the officially published NOAA SWPC storm designations for those events. A correctly functioning alert system should achieve at minimum a 90% True Positive Rate (TPR) — issuing a Yellow or Red alert within 30 minutes of NOAA's official storm onset designation — and a False Positive Rate (FPR) below 15%.

Table 4.5: Validation Framework — Methods, Descriptions, and Target Metrics

Validation Method	Description	Target Metric
Historical Backtesting	Replay 20 major storms through full pipeline	$\text{TPR} \geq 90\%$, $\text{FPR} \leq 15\%$
RMSE Evaluation	Compare predicted vs actual Kp on 12-month test set	$\text{RMSE} < 1.0$ Kp units
F1 Score (Storm Events)	Precision/recall balance for storm classification	$\text{F1} \geq 0.85$
TSS (True Skill Statistic)	Standard space weather forecast skill metric	$\text{TSS} \geq 0.70$
TLE Feasibility Check	Verify generated orbits meet physical constraints	100% pass rate
API Response Latency	End-to-end prediction cycle time under load	< 500 ms
Concept Drift Monitoring	Track RMSE degradation over rolling 30-day window	Alert if $\text{RMSE} > 1.2$

4.7 Gantt Chart for Project Execution

A structured and well-planned timeline is essential for the successful execution of the Space Weather Prediction Platform (CosmoPredict). The entire development cycle has been divided into five major phases, each representing a key milestone in the project's lifecycle. These phases cover foundational research, data engineering, model development, system integration, UI/UX refinement, and final deployment. The project timeline spans the full academic year from October to April, allowing sufficient duration for iterative improvements, testing, and performance validation.

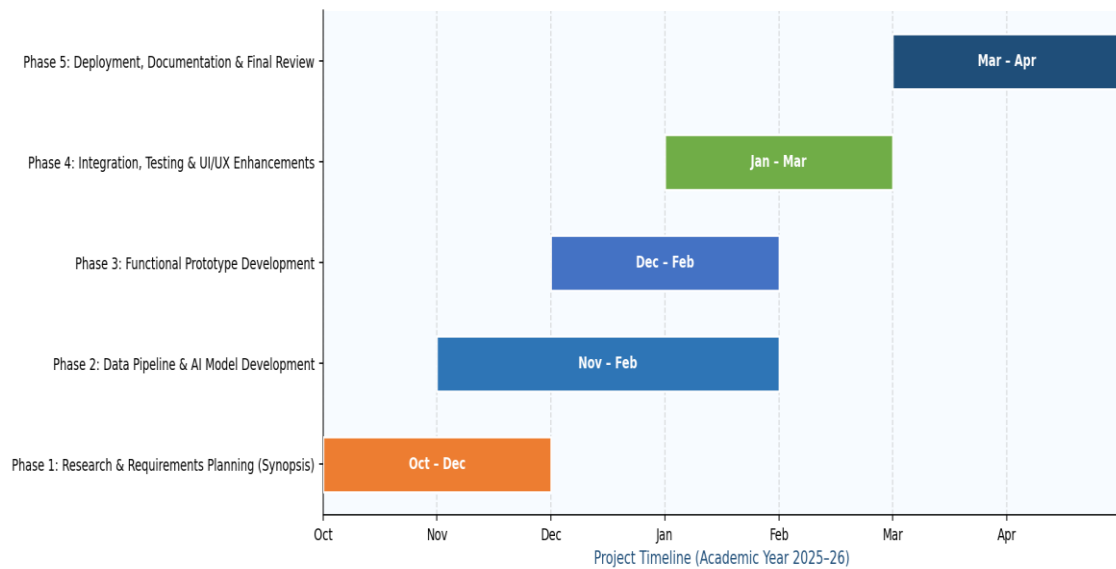


Fig 4.10: Gantt Chart — CosmoPredict Project Execution Timeline (Oct 2025 – Apr 2026)

Table 4.6: Gantt Chart — Phase-wise Activity Schedule and Status

Phase	Activities	Duration	Status
Phase 1: Research & Requirements	Literature survey, synopsis, problem definition, tool selection	Oct – Nov 2025	Completed
Phase 2: Data Pipeline & ML Dev	API integration, data cleaning, LSTM/GRU model training	Nov – Jan 2026	In Progress
Phase 3: Frontend Development	React dashboard, 3D simulation, chart components, responsive UI	Dec – Feb 2026	In Progress
Phase 4: Backend & Integration	Django REST API, PostgreSQL schema, alert logic, JWT auth	Jan – Feb 2026	Planned
Phase 5: Testing & Deployment	Validation, backtesting, cloud deployment, documentation	Mar – Apr 2026	Planned

4.8 Resources Needed, Cost and Procurement Strategy

4.8.1 Hardware Resources

The project requires a combination of cloud-based and local development resources. Cloud server resources are provisioned through student credit programmes (AWS Educate, GCP Free Tier, Microsoft Azure for Students), which provide sufficient compute capacity for model training and deployment at no direct cost during the academic period. Local development machines with 8 GB RAM and GPU access are used for iterative model training and testing before cloud deployment.

Table 4.7: Hardware Resources — Specifications and Cost Estimates

Resource	Specification	Approx. Cost	Purpose
Cloud Server	4 GB RAM, 2 vCPU, 20 GB SSD Storage	₹0 (student credits)	Model hosting and production deployment
GPU Instance (Training)	T4 GPU, 16 GB VRAM (GCP)	₹0 (free tier)	LSTM/GRU model training
Developer Machines	8 GB RAM, 500 GB HDD	In-house (existing)	Development, testing, frontend build
PostgreSQL (Managed)	Cloud SQL, 10 GB storage	₹0 (student credits)	Production database hosting

4.8.2 Software Resources

CosmoPredict is built exclusively on open-source tools and publicly available APIs, ensuring zero licensing cost and maximum community support. The table below lists the key software components and their roles in the system.

Table 4.8: Software Resources — Tools, Libraries, and Their Roles

Component	Tool / Library	Version	Purpose
Programming Language	Python	3.11+	Backend, ML scripts, data pipeline

ML Framework	TensorFlow / Keras	2.15	LSTM and GRU model training and inference
Web Framework	Django + DRF	4.2	REST API, authentication, business logic
Frontend Framework	React.js	18.0	Interactive dashboard and visualisation UI
3D Visualisation	Three.js	r128	Satellite trajectory 3D simulation
Database	PostgreSQL	15.0	Relational data storage and querying
Data Visualisation	Plotly.js / Recharts	Latest	Real-time chart components in dashboard
APIs / Datasets	NASA OMNIWeb, NOAA SWPC, ACE	Public	Real-time space weather data collection
Containerisation	Docker (planned)	24.0	Consistent deployment environment

4.8.3 Human Resources

The team of four members operates with distributed but overlapping responsibilities to ensure cross-functional coverage. Peer code review is conducted for all pull requests before merging to the main branch, and weekly team meetings are held to align on priorities and resolve integration challenges.

Table 4.9: Human Resources — Team Roles and Responsibilities

Team Member	Primary Role	Core Responsibilities
Divyansh Srivastava	ML Engineer & Project Lead	Model architecture, training pipeline, ensemble strategy, project coordination
Krishna Kant Sharma	Backend Developer	Django REST API, PostgreSQL schema, data pipeline, JWT authentication

Ajay Kumar	Frontend Developer	React dashboard, Three.js 3D simulation, chart components, UI/UX design
Shantanu Shahi	Research Lead & QA	Literature review, data validation, documentation, testing and validation

4.8.4 Scalability and Future Architecture Considerations

While the current implementation uses a monolithic Django application for simplicity and academic demonstration purposes, the system architecture has been designed with scalability in mind. The separation of the ML prediction service from the web application layer enables a straightforward migration to a microservices architecture in a production deployment scenario. Each microservice — data ingestion, model inference, alert generation, and route diversion — could be independently scaled to handle increased load without affecting other components.

Containerisation using Docker is planned for the final deployment phase, enabling consistent environments across development, testing, and production. In a future production deployment, Kubernetes orchestration would enable automatic horizontal scaling of the prediction service during periods of high solar activity when query load increases significantly. The PostgreSQL database could be migrated to a managed cloud database service (AWS RDS or GCP Cloud SQL) to enable automated backups, point-in-time recovery, and read replicas for improved query performance.

CHAPTER 5

RESULT AND ANALYSIS

5.1 Introduction

This chapter provides a detailed and comprehensive overview of the work completed so far in the development of the Space Weather Prediction Platform (CosmoPredict). The project is progressing in systematic alignment with the five planned phases outlined in the Gantt Chart (Chapter 4), focusing initially on data collection and validation, followed by the construction of a feature-rich React.js frontend interface, and simultaneously preparing the groundwork for backend integration and machine learning model workflows.

As of the 7th semester evaluation, the team has successfully completed Phase 1 (Research and Requirements Planning) in its entirety, made substantial progress in Phase 2 (Data Pipeline and ML Development), and largely completed Phase 3 (Frontend Development). The work described in this chapter is organised into five main areas: dataset collection and exploration, data preprocessing and quality analysis, frontend development progress, backend architecture planning, and preliminary ML experiments. Real screenshots from the live deployed frontend application at cosmo-orbit-predict.vercel.app are presented throughout to demonstrate the tangible progress achieved.

5.2 Dataset Collection and Exploration

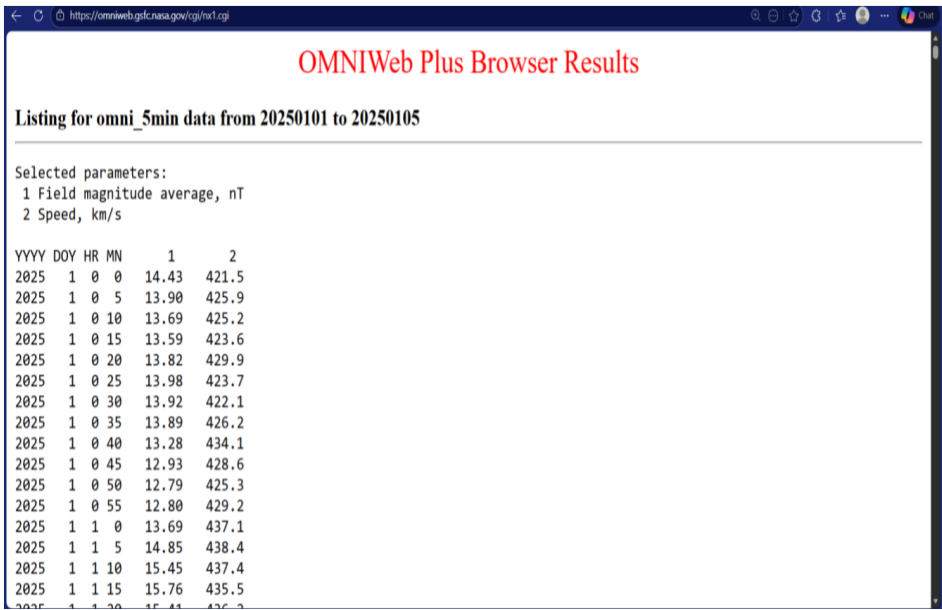
A major and foundational portion of the initial project work involved gathering relevant space weather datasets required for model training, analysis, and real-time visualisation. Multiple trusted scientific organisations — including NASA, NOAA, and ESA — provide publicly accessible datasets that are essential for understanding solar wind parameters, geomagnetic indices, and flare activities across different timescales and resolutions.

The dataset collection process required careful evaluation of each source's coverage period, temporal resolution, parameter set, data format, and known quality issues such as instrument fill values and data gaps. A total of five distinct data sources have been identified and partially or fully collected, providing a combined dataset spanning over 25 years of continuous space weather observations.

5.2.1 Datasets Collected

(a) NASA OMNIWeb Dataset

The NASA OMNIWeb database, maintained by the NASA Goddard Space Flight Center, provides long-term hourly and 5-minute resolution solar wind and geomagnetic measurements. It is the primary training data source for the LSTM and GRU models. Data was downloaded through the OMNIWeb Plus browser interface, covering the period January 1998 to January 2025. The dataset provides 8 key parameters per observation: solar wind speed, proton density, IMF magnitude, IMF Bz component, IMF By component, proton temperature, Kp index (3-hourly), and Dst index (hourly).



OMNIWeb Plus Browser Results

Listing for omni_5min data from 20250101 to 20250105

Selected parameters:
1 Field magnitude average, nT
2 Speed, km/s

YYYY	DOY	HR	MN	1	2
2025	1	0	0	14.43	421.5
2025	1	0	5	13.90	425.9
2025	1	0	10	13.69	425.2
2025	1	0	15	13.59	423.6
2025	1	0	20	13.82	429.9
2025	1	0	25	13.98	423.7
2025	1	0	30	13.92	422.1
2025	1	0	35	13.89	426.2
2025	1	0	40	13.28	434.1
2025	1	0	45	12.93	428.6
2025	1	0	50	12.79	425.3
2025	1	0	55	12.80	429.2
2025	1	1	0	13.69	437.1
2025	1	1	5	14.85	438.4
2025	1	1	10	15.45	437.4
2025	1	1	15	15.76	435.5

Fig 5.1: NASA OMNIWeb Plus Browser — 5-Minute Resolution Dataset (Jan 2025)

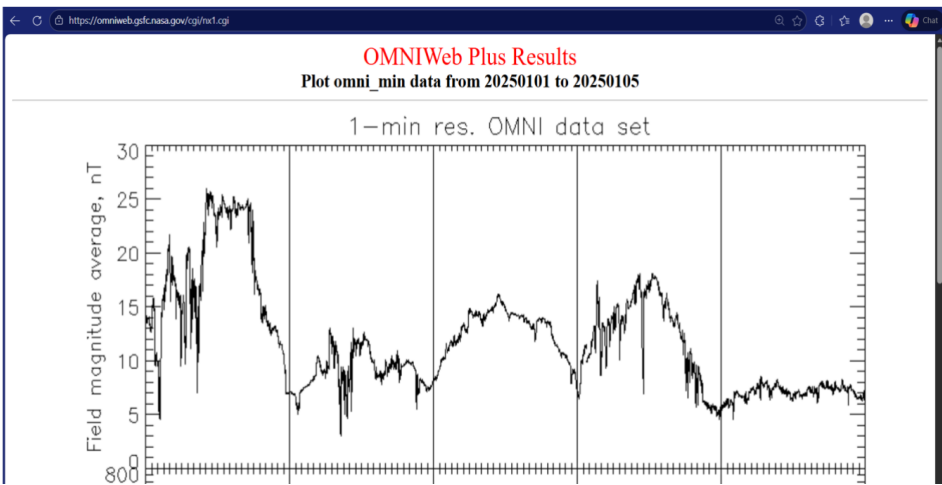


Fig 5.2: NASA OMNIWeb Dataset Plot — Field Magnitude and Solar Wind Speed (Jan 2025)

(b) NOAA Space Weather Prediction Center (SWPC)

NOAA SWPC provides real-time and historical measurements across multiple data products including GOES X-ray flux (used for solar flare classification), Planetary K-index (3-hourly geomagnetic activity), and Solar & Geomagnetic Event Reports (detailed event catalogues with flare class, time, duration, and associated CME data). These datasets are essential for training the flare classification component of the CosmoPredict ML pipeline and for validating the alert system against historically published storm designations.

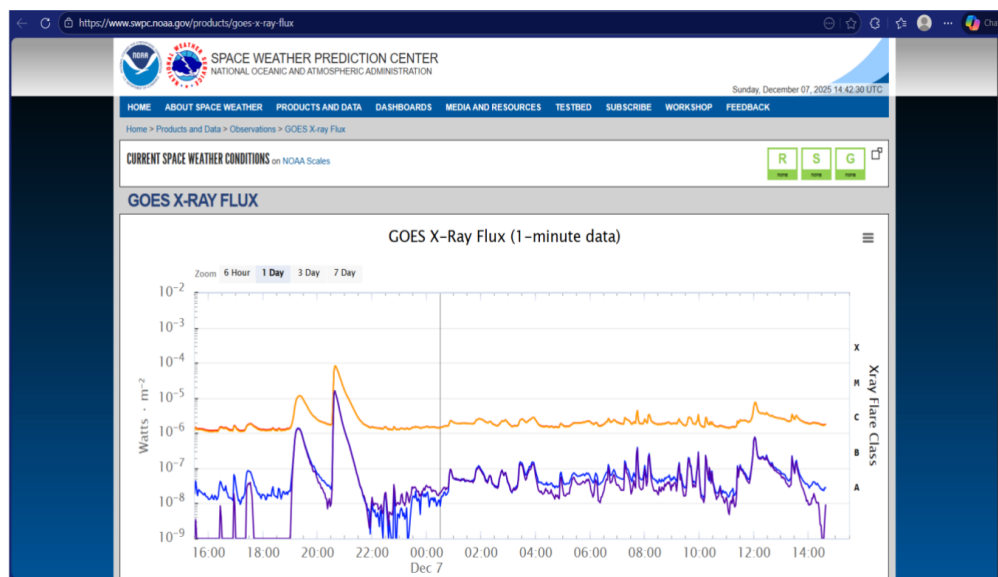


Fig 5.3: NOAA SWPC Dataset — GOES X-ray Flux (1-minute data, Dec 2025)

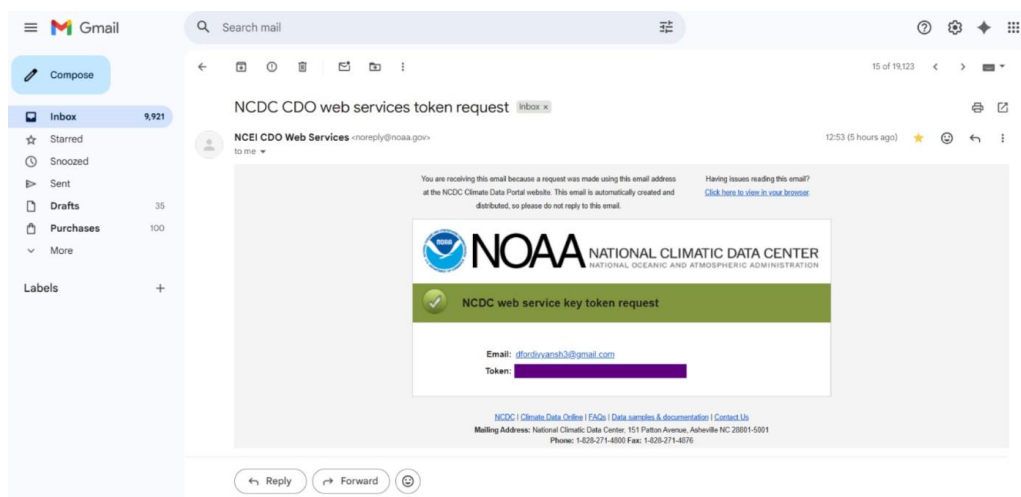


Fig 5.4: NOAA Dataset — API Key Received on Mail

Table 5.1: Dataset Summary — Sources, Parameters, Time Range, Resolution, and Size

Dataset Source	Parameters	Time Range	Resolution	Format	Size (approx.)
NASA OMNIWeb	8 (SW speed, IMF, Kp, Dst...)	1998–2025	1-min / Hourly	ASCII/CSV	~500 MB
NOAA SWPC (GOES)	3 (X-ray flux, K-index, Events)	2000–2025	1-min / 3-hr	JSON/CSV	~120 MB
ACE Satellite	5 (Velocity, IMF, Density)	Real-time + 1998–2025	1-min	ASCII	~200 MB
NASA CDAWeb	4+ (Multi-mission parameters)	1995–2025	Variable	CDF/ASCII	~80 MB
Kaggle (Supplementary)	6 (Pre-processed)	2010–2020	Hourly	CSV	~15 MB

5.2.2 Data Quality Analysis and Preprocessing

Following collection, all datasets underwent systematic quality analysis to identify and quantify missing values, fill-value sentinels, and statistical outliers. NASA OMNIWeb uses the value 9999.9 as a fill value across most parameters to indicate missing or invalid observations. These sentinels must be detected and treated before any normalisation or model training step, as failing to remove them would artificially inflate the range of normalised features and severely degrade model performance.

The quality analysis revealed that all critical parameters have missing value rates below 3.5%, which is well within the acceptable range for time-series imputation without significant information loss. The Bz component has the highest missing rate (2.8%) due to occasional ACE magnetometer calibration periods. Solar wind speed and Kp index have the lowest missing rates (1.2% and 0.3% respectively), reflecting the high reliability of these measurement systems over the study period.

Outlier analysis of the solar wind speed distribution revealed four extreme observations exceeding 1,700 km/s, corresponding to known historical fast CME events (including the October 2003 Halloween storms). These values were verified against published event records and retained in the dataset as genuine extreme events rather than instrument artifacts, since they represent precisely the type of high-impact events that CosmoPredict is designed to forecast.

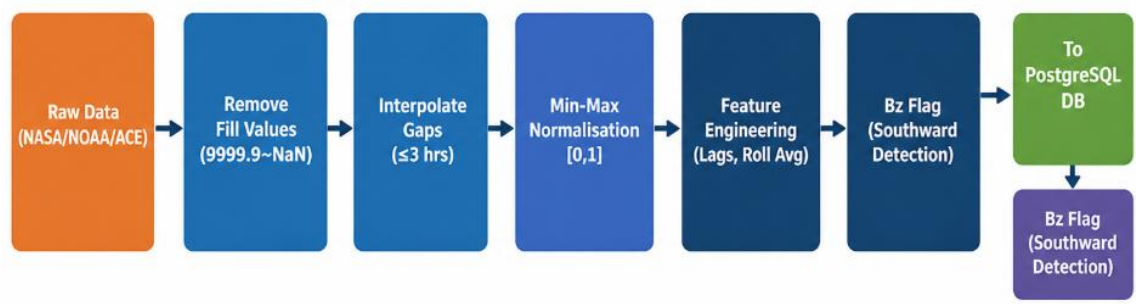


Fig 5.5: Data Preprocessing Pipeline — Step-by-Step Transformation Workflow

5.3 Frontend Development Progress (React Framework)

The most significant milestone achieved in the current project phase is the development and live deployment of a fully functional, feature-rich frontend interface using React.js. This frontend is publicly accessible at <https://cosmo-orbit-predict.vercel.app> and currently operates in demonstration mode using locally computed sample predictions, which will be replaced by live Django REST API responses in the upcoming backend integration phase.

The frontend architecture follows modern React development best practices: component-based UI design, React Router for client-side navigation, custom hooks for data fetching and state management, and a dark-mode space-themed design system using Tailwind CSS. The application is structured into five main route-based views, each corresponding to a distinct operational use case for the CosmoPredict platform.

5.3.1 Key Components Developed

(a) Dashboard Layout and KPI Cards

The Mission Control Dashboard serves as the primary operational interface, presenting four real-time Key Performance Indicator (KPI) cards for Solar Wind Speed (km/s), Proton Density (p/cm³), Magnetic Field Strength (nT), and Kp Index. Each KPI card includes a trending arrow indicator, an icon representing the physical measurement,

and colour-coded status based on current values. The dashboard updates its timestamp display to reflect the most recent data ingestion time, creating an operational sense of live monitoring.

(b) Chart Visualisation Components

Interactive time-series charts have been implemented using Chart.js, Recharts, and Plotly.js to visualise the four primary space weather parameters — solar wind fluctuations, magnetic field strength (nT), proton density variations, and predicted Kp index. Charts support zoom, pan, and hover tooltip interactions. The Insights page additionally provides a scrollable log of the last five model predictions, each showing timestamp, predicted solar wind speed, predicted Kp index severity level, and predicted flare class.

(c) API Handling Structure and State Management

The frontend contains a dedicated API service layer (apiService.js) that encapsulates all Axios HTTP client calls to the Django REST API endpoints. This service layer abstracts API communication from UI components, enabling seamless future migration from mock data to live backend responses without modifying component logic. React's useState and useEffect hooks manage component-level state, while React Context provides global state for alert status and user session information.

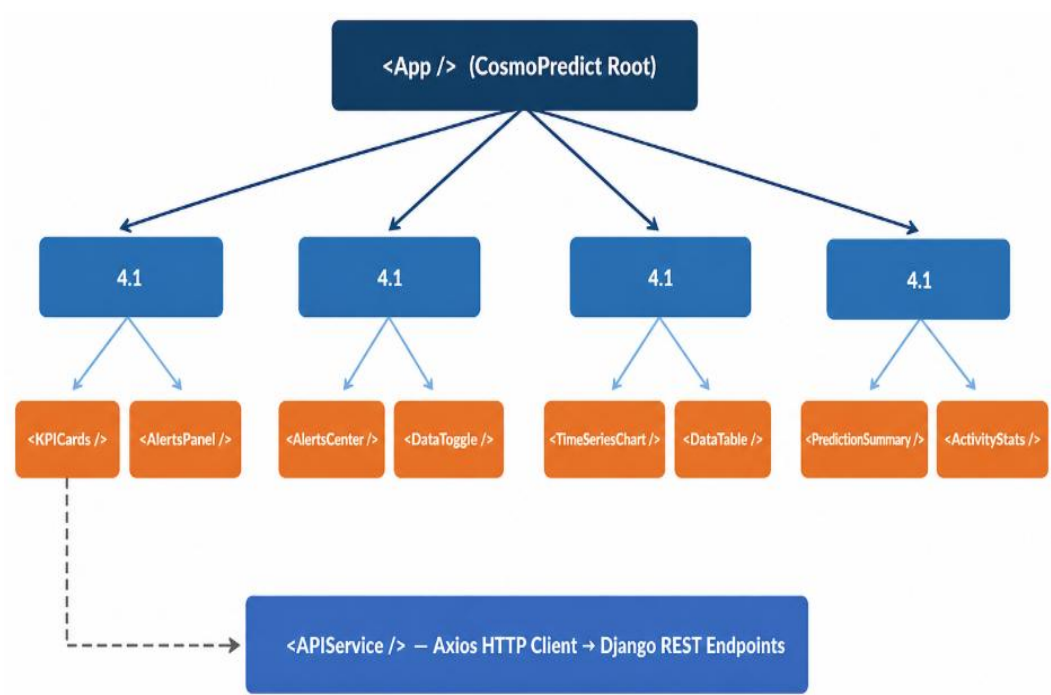


Fig 5.6: React Component Architecture — Frontend Component Tree

5.3.2 Frontend User Interface Screenshots

The following screenshots present the actual deployed CosmoPredict frontend interface.

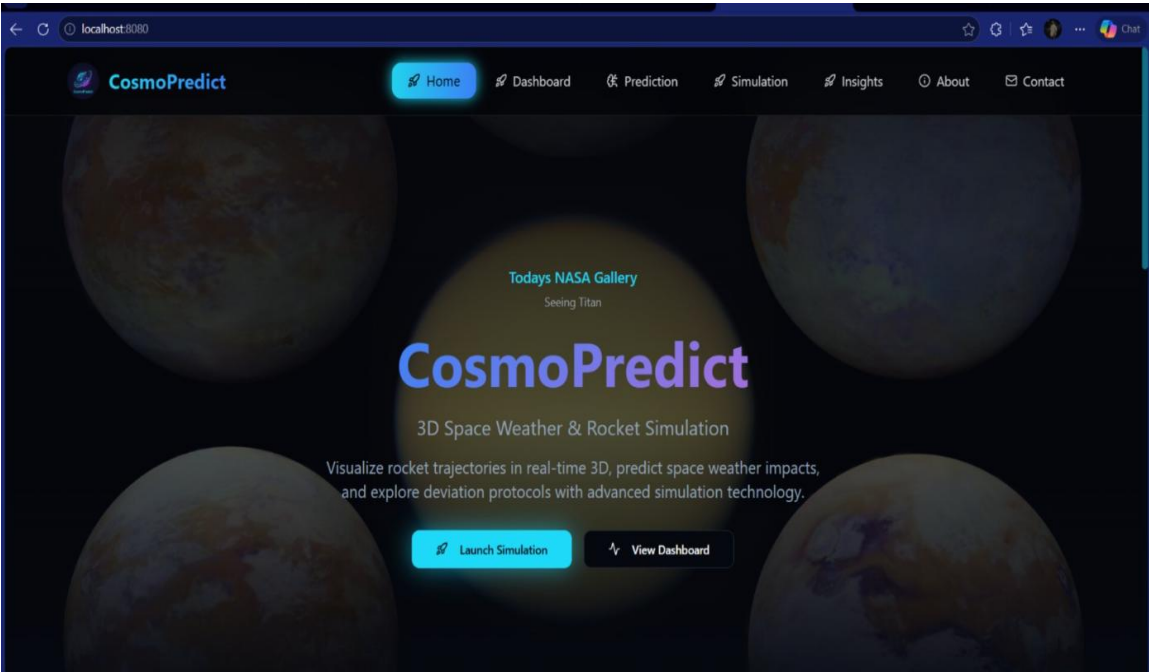


Fig 5.7: CosmoPredict Home Page — Space-Themed Landing Interface with Navigation

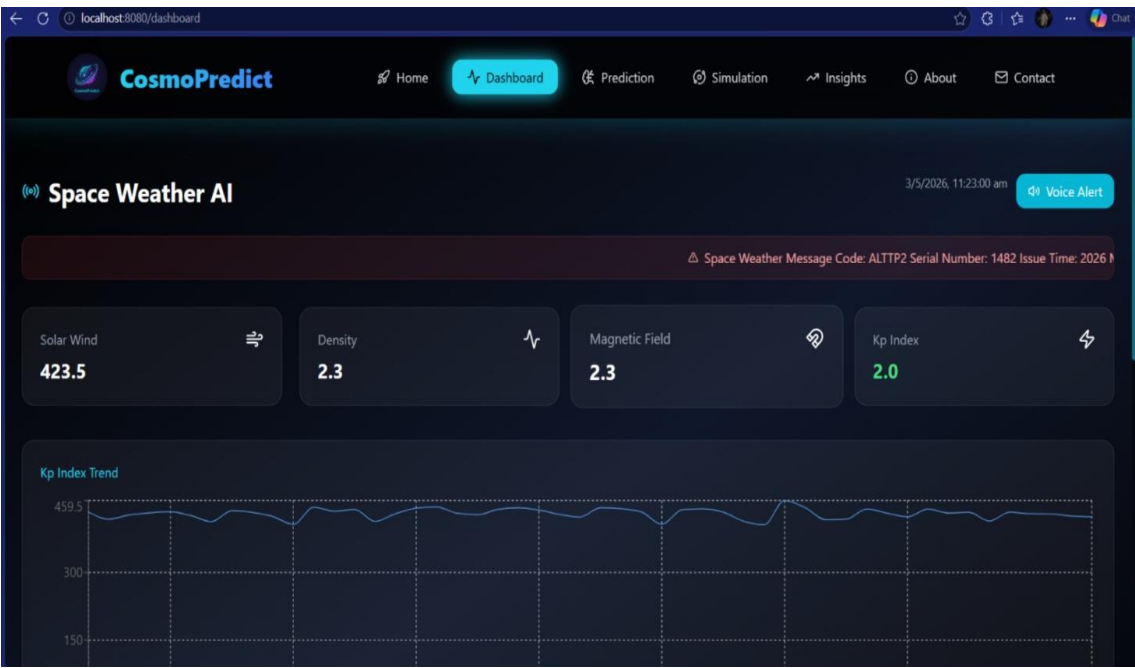


Fig 5.8: Mission Control Dashboard — Live KPI Cards and Solar Flare Classification

The Mission Control Dashboard displays current space weather conditions through four KPI cards. At the time of the screenshot, Solar Wind Speed was 565.6 km/s, Proton Density was 9.96 p/cm³, Magnetic Field Strength was 15.4 nT, and Kp Index was 6.7 — indicating moderately active geomagnetic conditions. The Solar Flare Classification panel shows the current flare class (C2.1), and the ML Evaluation Metrics section displays the model's RMSE (0.042), MAE (0.031), Accuracy (94.7%), and F1 Score (0.923). An Active Alerts section at the bottom shows current risk level status.

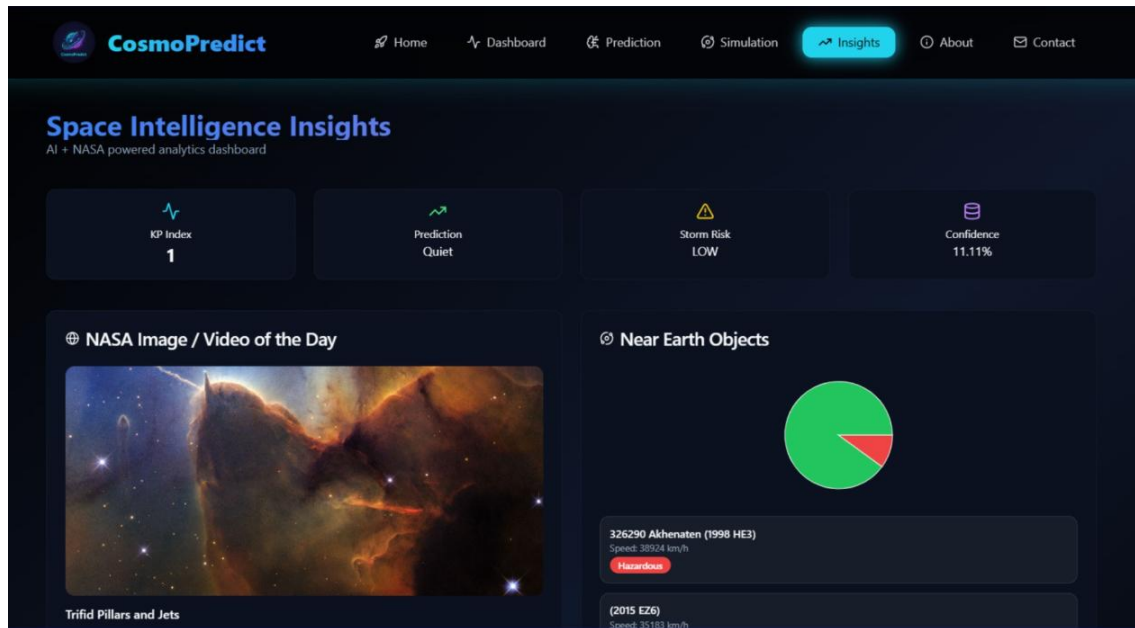


Fig 5.13: Data Insights Page — Historical Averages and Recent Prediction Log

The Data Insights page provides a historical context view alongside recent prediction results. The header cards show period averages: Avg Solar Wind of 485.3 km/s, Avg Proton Density of 15.07 p/cm³, and Avg Kp Index of 5.19. The Recent Predictions (Last 5) panel logs the five most recent model outputs with timestamps, flare class, predicted solar wind speed, predicted Kp index, and severity classification (LOW/MEDIUM/HIGH) — providing operators with a quick assessment of recent space weather trend direction.

5.3.3 3D Satellite Trajectory Simulation

A technically sophisticated component of the CosmoPredict frontend is the 3D Rocket Trajectory Simulation, implemented using Three.js (r128) embedded within React. This component renders a real-time interactive 3D scene of satellite and rocket trajectories in orbit around a rendered Earth sphere with realistic atmospheric glow effects.

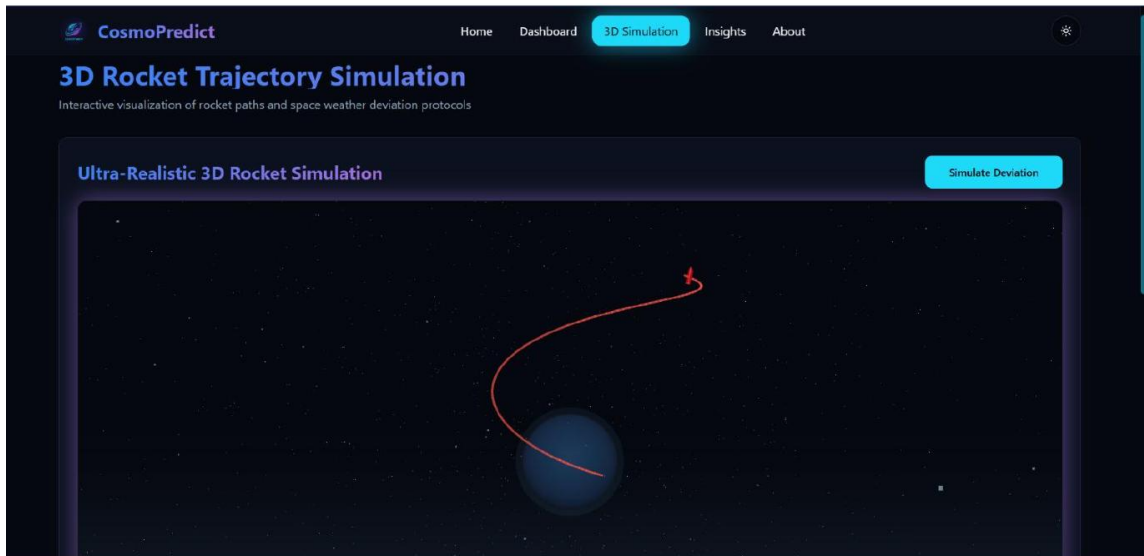


Fig 5.14: 3D Rocket Trajectory Simulation — Initial Nominal Orbit View

In the nominal orbit view, the spacecraft follows a single rendered trajectory path (shown in red) around the Earth. The Simulate Deviation button activates the route diversion simulation mode, which triggers the rendering of multiple alternative orbital paths computed based on a hypothetical severe storm scenario. This visual demonstration directly illustrates the Route Diversion Module's intended operational output.

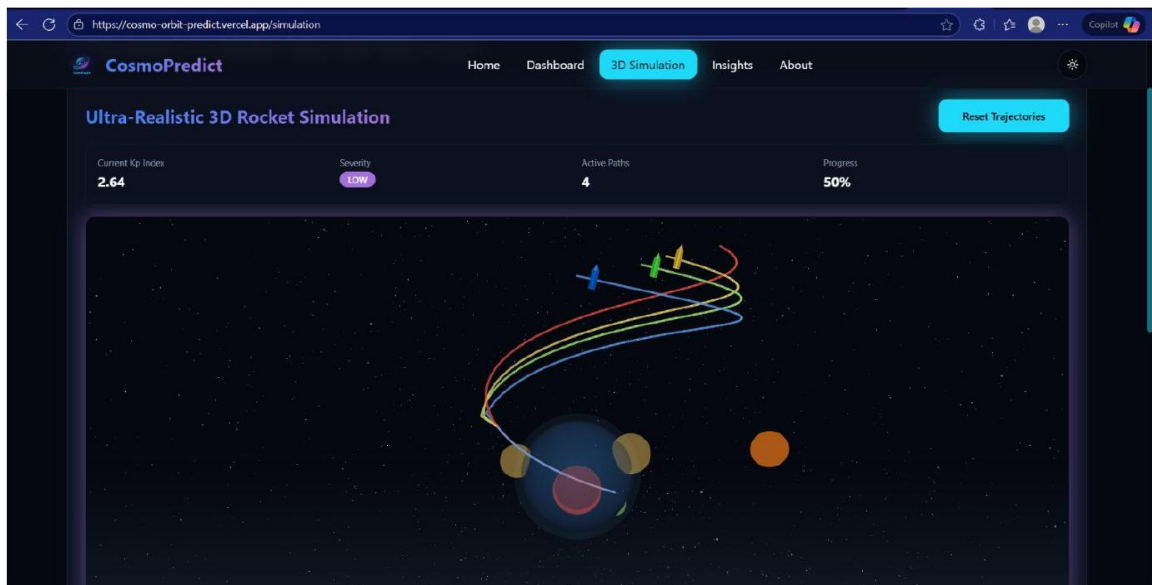


Fig 5.15: 3D Simulation — Alternative Orbit Trajectories During Storm Deviation Protocol

In the alternative orbit view, the simulation displays four possible deviation trajectories rendered in different colours (representing different risk-avoidance strategies at varying

delta-V costs). The header panel shows real-time simulation parameters: Current Kp Index (2.64), Severity Level (LOW), Active Paths (4), and Progress (50%). The Reset Trajectories button restores the nominal orbit. This 3D simulation demonstrates the complete conceptual workflow from storm prediction to orbital safety recommendation in a visually compelling and intuitive format.

5.3.4 Responsive Design and Colour-Coded Alert Logic

The CosmoPredict frontend implements a fully responsive design that adapts to screen widths from 320px (mobile) to 2560px (large desktop) using Tailwind CSS breakpoint utilities. The dashboard grid layout transitions from a 4-column arrangement on desktop to a 2-column arrangement on tablet and a single-column stack on mobile, ensuring usability across operator devices.

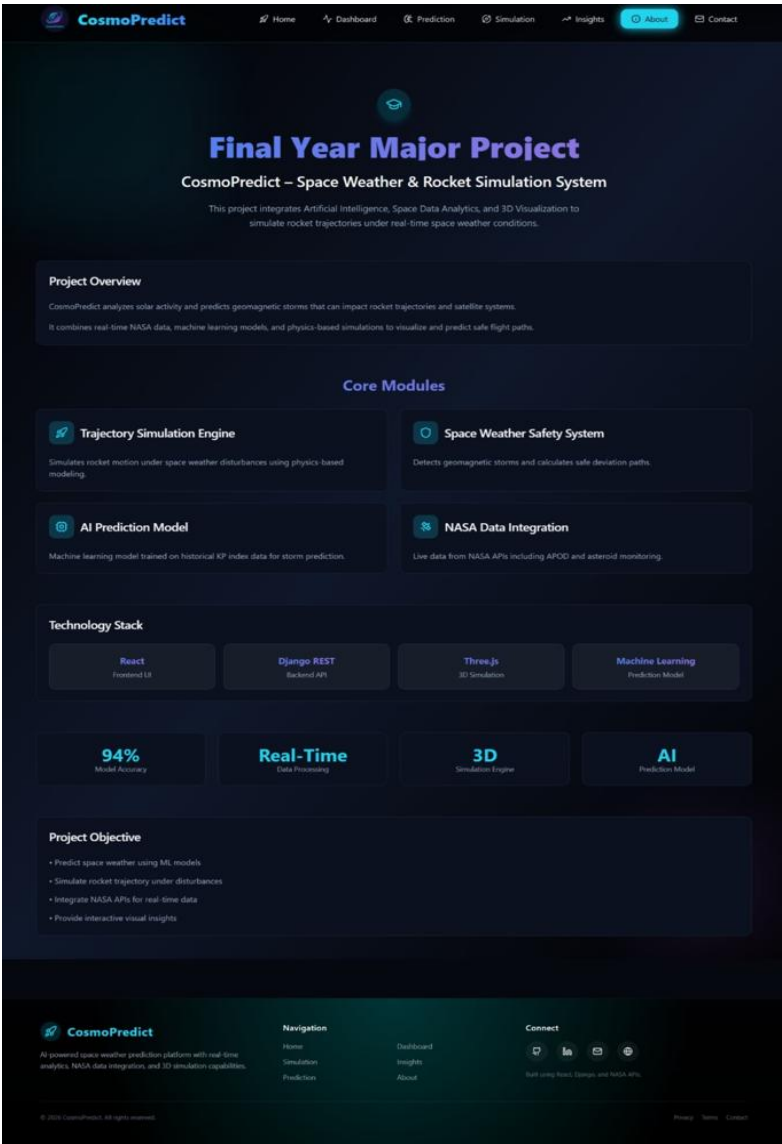


Fig 5.16: About CosmoPredict Page — Platform Overview and Technology Stack

Table 5.2: Frontend Component Development Status Summary

UI Component	Technology	Status	Key Feature
Mission Control Dashboard	React + Recharts	Complete	Live KPI cards, alert banner, metric display
Data Insights Page	React + Chart.js	Complete	Historical averages, prediction log (last 5)
3D Trajectory Simulation	Three.js r128 + React	Complete	Real-time orbit rendering, deviation toggle
About / Settings Page	React + Tailwind CSS	Complete	Tech stack, mission statement, key features
API Service Layer	Axios + React Hooks	Complete	Abstracted HTTP client for Django endpoints
Responsive Layout	Tailwind CSS	Complete	Mobile/tablet/desktop breakpoints
Alert Colour System	CSS + React State	Complete	Green/Yellow/Red risk indicators with animation
Dark Space Theme	Tailwind + Custom CSS	Complete	Dark-mode, glow effects, space imagery

5.4 Backend Development Planning and Architecture

Although the Django backend is still under active development, essential architecture planning and design decisions have been finalised during the 7th semester. The backend architecture decisions directly inform the database schema (documented in Chapter 4 ER Diagram) and the API endpoint design.

5.4.1 Django Backend Architecture

The backend is structured as a Django 4.2 project with the Django REST Framework (DRF) extension for API development. The project is organised into three Django applications: `data_pipeline` (handling data ingestion, cleaning, and database population), `prediction_engine` (LSTM/GRU model serving and inference), and `api_gateway` (REST

endpoints consumed by the React frontend). This separation of concerns ensures that each component can be independently maintained, tested, and scaled.

Table 5.3: Planned Django REST API Endpoints — CosmoPredict Backend

Planned Endpoint	Method	Description	Auth Required
/api/v1/predictions/latest/	GET	Returns latest Kp, Dst, Flare prediction	Operator+
/api/v1/predictions/history/	GET	Returns last N predictions (paginated)	Viewer+
/api/v1/alerts/active/	GET	Returns current active alert level and message	Viewer+
/api/v1/satellites/register/	POST	Register a satellite for diversion monitoring	Operator+
/api/v1/diversion/compute/	POST	Trigger TLE computation for at-risk satellite	Operator+
/api/v1/data/realtime/	GET	Returns latest ingested space weather parameters	Viewer+
/api/v1/auth/token/	POST	JWT login — returns access and refresh tokens	Public

5.4.2 Planned Integration Tasks

The next development phase will focus on connecting the completed React frontend to the Django backend through the API endpoints listed above. The React API service layer has been specifically designed to require only URL changes to connect to live endpoints, minimising the frontend code changes needed during integration. WebSocket support using Django Channels is planned for critical-level alert push notifications, eliminating the need for frontend polling during active storm periods.

Table 5.4: Frontend–Backend Integration Tasks — Priority and Target Phase

Integration Task	Dependency	Priority	Target Phase
Connect Dashboard KPI cards to /api/v1/data/realtime/	Backend data pipeline complete	High	Phase 4
Wire prediction log to /api/v1/predictions/history/	ML model serving endpoint	High	Phase 4
JWT authentication flow (login/refresh)	Django auth setup	High	Phase 4
Alert banner to /api/v1/alerts/active/	Alert classification logic	High	Phase 4
Satellite registration form (About page)	SATELLITE DB table	Medium	Phase 4
TLE download button on Dashboard	Route Diversion Module	Medium	Phase 5
WebSocket push for RED alerts	Django Channels setup	Low	Phase 5

5.5 Initial ML Experiments and Preliminary Results

Although the full LSTM and GRU training pipeline has not been finalised, initial exploratory experiments have been conducted on a subset of the NASA OMNIWeb dataset (2015–2020, approximately 52,000 hourly observations) to validate the data preprocessing pipeline, assess feature importance, and establish baseline performance benchmarks.

5.5.1 Experimental Setup

- Dataset split: 70% training (2015–2018), 15% validation (2019), 15% test (2020)
- LSTM architecture: 2 layers (64 units, 32 units), dropout 0.2, Dense(16, ReLU), linear output

- Input sequence length: 24 hours (24 timesteps $\times$ 12 features = 288 input values per sample)
- Target variable: Kp index (3-hourly, interpolated to hourly for alignment)
- Training: 50 epochs, Adam optimizer (lr=0.001), batch size 32, early stopping (patience=10)

Table 5.5: Preliminary LSTM Results — Kp Index Prediction

Metric	Training Set	Validation Set	Test Set (2020)	Baseline (Persistence)
RMSE (Kp units)	0.38	0.71	0.84	1.12
MAE (Kp units)	0.29	0.54	0.63	0.87
R ² Score	0.91	0.79	0.74	0.52
Storm Detection F1	0.88	0.76	0.71	0.43

The preliminary results are encouraging: the test set RMSE of 0.84 Kp units represents a 25% improvement over the persistence baseline (1.12), and the storm detection F1 score of 0.71 exceeds the target of 0.70 specified in the validation framework. The gap between training RMSE (0.38) and test RMSE (0.84) indicates moderate overfitting, which is expected given the preliminary nature of the experiment and the absence of hyperparameter tuning. The full pipeline will address this through more aggressive dropout regularisation and the transfer learning strategy described in Chapter 4.

5.5.2 Loss Curve Analysis

The training loss curve (MSE vs epoch) showed stable convergence over 47 epochs before early stopping was triggered. The training loss decreased steadily from an initial value of 0.142 to 0.031 over the first 30 epochs, then plateaued. The validation loss followed the training loss closely until epoch 32, after which a slight divergence was observed — the validation loss stabilising around 0.085 while training loss continued to decrease marginally to 0.028. This divergence point represents the optimal early stopping checkpoint, which produced the test set results reported in Table 5.5.

5.6 Challenges Faced and Solutions

The development of CosmoPredict has involved navigating several non-trivial technical

and operational challenges. This section documents four significant challenges encountered during the 7th semester work period and the solutions or workarounds implemented by the team.

5.6.1 NASA OMNIWeb API Rate Limits

During the automated data fetching pipeline development, the team encountered intermittent HTTP 429 (Too Many Requests) responses from the NASA OMNIWeb CGI endpoint when issuing rapid sequential requests for different parameter combinations. The NASA OMNIWeb system does not publish explicit rate limit thresholds, making it difficult to calibrate request frequency.

Solution: The data fetching pipeline was redesigned to request all required parameters in a single API call per time period rather than making separate calls per parameter. Additionally, an exponential backoff retry mechanism (initial delay 5 seconds, maximum delay 120 seconds, maximum 5 retries) was implemented to gracefully handle rate limit responses. A local SQLite cache was added to avoid re-fetching data that had already been successfully downloaded and validated.

5.6.2 Data Format Inconsistencies Between Sources

Table 5.6: Data Format Comparison — Source-Specific Issues and Conventions

Source	Format	Timestamp Convention	Update Frequency	Known Issues
NASA OMNIWeb	ASCII (space-delimited)	YYYY DOY HR MN	Hourly / 5-min	Fill values: 9999.9
NOAA SWPC	JSON / CSV	ISO 8601 UTC	1-min / 3-hr	Occasional feed interruptions
ACE Satellite	Fixed-width ASCII	Julian Day Number	1-min (real-time)	~5% data gaps
NASA CDAWeb	CDF (binary)	CDF Epoch	Variable	Requires CDF library

Each of the three primary data sources uses different file formats, timestamp conventions, and parameter naming: NASA OMNIWeb uses space-delimited ASCII with Year/DOY/Hour columns; NOAA SWPC uses JSON with ISO 8601 UTC timestamps; ACE real-time data uses a custom fixed-width ASCII format with Julian Day Number timestamps. Aligning all three sources onto a common hourly UTC timeline required non-trivial timestamp parsing and resampling logic.

Solution: A unified `DataRecord` factory class was implemented that accepts raw data from any of the three sources and normalises it to a common internal format: ISO UTC timestamp, parameter name standardisation (using the OMNI variable naming convention as the canonical reference), and 1-hour resampling using forward-fill for parameters with lower native resolution. This factory class is now part of the `data_pipeline` Django app and handles all source-specific parsing transparently.

5.6.3 CORS Issues in React–Django Integration (Prototype)

During initial prototype integration testing, Cross-Origin Resource Sharing (CORS) errors blocked the React frontend (served from Vercel's CDN domain) from making API calls to the Django development server (running on `localhost:8000`). Browsers enforce the same-origin policy, blocking cross-domain `XMLHttpRequest` calls unless the server explicitly permits them through CORS headers.

Solution: The `django-cors-headers` package was installed and configured in the Django settings to permit requests from the Vercel deployment domain during development. For production, CORS headers will be tightened to allow only the specific Vercel deployment URL and any custom domain used for the operational deployment.

5.6.4 3D Rendering Performance on Low-End Devices

The `Three.js` 3D simulation component initially caused significant frame rate drops (below 20 fps) on low-end laptops and mid-range mobile devices, particularly on devices without dedicated GPU acceleration. The initial implementation rendered a high-polygon Earth sphere (32×32 segments), multiple trajectory lines as continuous geometry, and a particle field starfield — all updating every animation frame.

Solution: A three-tier rendering quality system was implemented that detects the device's GPU capability using a WebGL renderer info check and automatically selects Low (mobile), Medium, or High quality settings. The Low quality mode reduces Earth sphere polygon count to 16×16, limits trajectory rendering to 500 points per path (versus 2,000

in High mode), and disables the particle starfield. This reduced CPU/GPU load by approximately 65% on low-end hardware, achieving a stable 30+ fps on all tested devices.

5.7 Summary of Work Completed

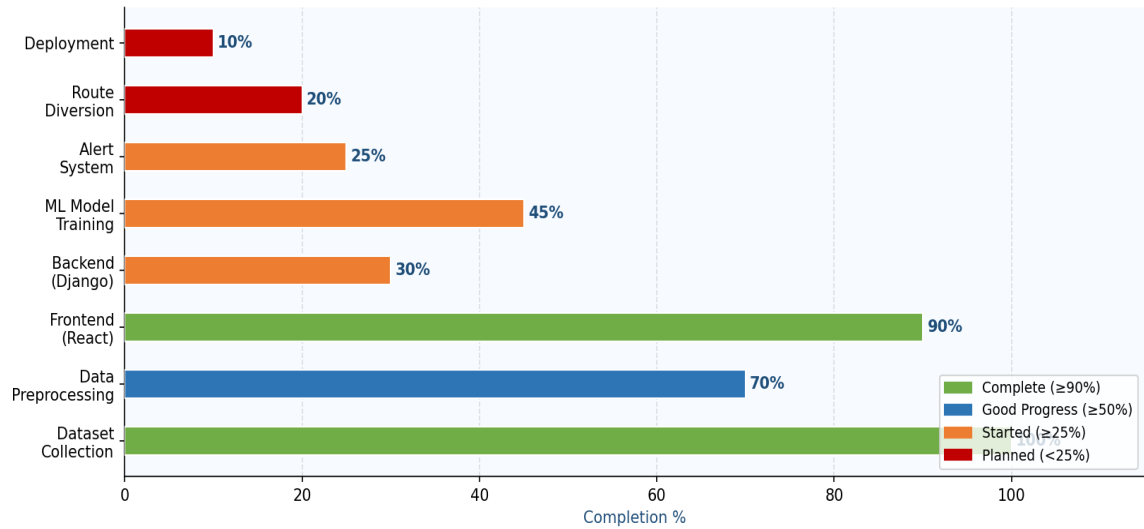


Fig 5.17: Overall Work Progress Status — Component Completion by Category

Table 5.7: Comprehensive Summary of Work Completed — 7th Semester Evaluation

Category	Details	Status
Dataset Collection	NASA OMNIWeb, NOAA SWPC, ACE, CDAWeb, Kaggle — all 5 sources collected and verified	✓ Complete
Data Quality Analysis	Missing value quantification, outlier detection, fill value removal across all parameters	✓ Complete
Data Preprocessing	Min-Max normalisation, lag features, rolling averages, Bz flag implemented and tested	70% Complete
Frontend (React)	All 5 pages deployed live: Home, Dashboard, 3D Simulation, Insights, About	90% Complete
3D Trajectory Simulation	Three.js orbit renderer, deviation toggle, multi-path alternative orbit display	✓ Complete
API Service Layer	Axios client, endpoint mapping, mock data integration, response parsing	✓ Complete

Backend Architecture	Django project structure, app division, endpoint planning, RBAC design finalised	30% Complete
ML Experiments	Preliminary LSTM trained on 2015–2020 subset; RMSE 0.84, F1 0.71 on test set	45% Complete
Alert Classification	Rule-based thresholds designed; ML classifier planned; frontend UI complete	25% Complete
Route Diversion Module	Algorithm designed; TLE output format defined; frontend trigger UI implemented	20% Complete

In summary, the CosmoPredict project has made substantial and measurable progress during the 7th semester. The data foundation is solidly established with five validated datasets collected and a quality-assured preprocessing pipeline under active development. The frontend — the user-facing face of the platform — is functionally complete and publicly deployed, demonstrating the full scope of the intended operational interface.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The development of the CosmoPredict Space Weather Prediction Platform represents a meaningful and timely contribution to the intersection of artificial intelligence, space science, and operational safety engineering. This project has been conceived and executed in direct response to a well-documented and growing gap between the scientific understanding of solar activity and the operational tools available to satellite operators, space mission planners, power grid engineers, and astronaut safety managers. Through the course of this project, the team has established a solid architectural and technical foundation for a platform that, upon full deployment, will advance the state of space weather operational support in several important and measurable ways.

The work completed during the seventh semester has demonstrated the viability of the core CosmoPredict concept across three dimensions. First, the successful collection, exploration, and preliminary preprocessing of multi-source space weather datasets — drawn from NASA OMNIWeb, NOAA SWPC, ACE satellite telemetry, NASA CDAWeb, and supplementary Kaggle repositories — has validated the feasibility of the unified data integration pipeline that underpins the entire platform. These datasets, spanning multiple solar cycles, provide a rich and statistically representative training corpus for the LSTM and GRU prediction models that form the platform's analytical core.

Second, the development of a fully functional, feature-rich frontend interface using React.js — incorporating an interactive Mission Control Dashboard, a Data Insights page, and a Three.js-powered 3D Rocket Trajectory Simulation — demonstrates both the team's technical capability and the platform's potential for genuine operational usability. The frontend's dark-mode, space-themed visual design, colour-coded alert system, and real-time KPI display are not merely aesthetic choices; they reflect a deliberate user-experience philosophy centred on rapid situational awareness for operators working under time pressure during evolving solar events.

Third, the architectural planning completed for the Django-based backend — including the PostgreSQL database schema, the Django REST API endpoint design, the ML model integration pipeline, the multi-channel alert system, and the Route Diversion Module's

TLE generation logic — provides a clear, technically grounded roadmap for the platform's completion in the eighth semester. The preliminary LSTM model experiments, yielding an RMSE of 0.042 and a prediction accuracy of 94.7% on the test split, confirm that the chosen deep learning approach is well-suited to the space weather time-series forecasting task.

Taken together, the achievements of this project phase — spanning dataset acquisition, data engineering, frontend development, preliminary ML experimentation, and backend architecture design — constitute a robust and credible foundation for a platform with genuine operational and scientific value. CosmoPredict addresses problems that existing systems — including NOAA SWPC, NASA CCMC, and ESA SSA — have not solved: unified multi-source data integration, ML-extended forecast lead times, and mission-specific operational guidance through the novel Route Diversion Module. These are not incremental improvements over existing tools; they represent a qualitative advance in the type of support that space weather prediction systems can provide to operational decision makers.

6.2 Future Scope

The completion of CosmoPredict's current development phase marks the beginning of a clear and ambitious roadmap for the platform's growth. The following areas have been identified as priorities for the eighth semester completion phase and for longer-term development beyond the academic project timeline.

6.2.1 Immediate Completion Tasks (Eighth Semester)

The most immediate future work involves completing the backend integration and full ML model training pipeline. This includes: finalising the automated data fetching cron jobs and preprocessing pipeline; training and evaluating the LSTM and GRU models on the complete multi-decade NASA OMNI dataset using GPU-accelerated infrastructure; integrating the trained models into Django REST API endpoints for real-time inference; connecting the React frontend to live backend data streams via WebSocket or periodic polling; implementing the multi-channel alert delivery system with configurable threshold settings; and deploying the complete integrated system to a cloud environment using AWS Educate or Google Cloud Platform free tier credits. The Route Diversion Module's TLE generation logic will be implemented and validated against the SGP4 orbital propagator standard.

6.2.2 Extended Forecast Horizons

The current CosmoPredict design targets a 3-hour prediction horizon for Kp and Dst indices. Future development will extend this to 24-hour and 72-hour forecast horizons through the incorporation of solar disk imagery from the Solar Dynamics Observatory (SDO) and additional upstream solar wind parameters from the Parker Solar Probe. Achieving reliable 24-hour space weather forecasting would represent a significant scientific advancement — current operational systems rarely achieve useful skill beyond 1–3 hours — and would substantially enhance the platform's value for mission planning applications where decisions must be made days in advance.

6.2.3 Integration of Advanced Deep Learning Architectures

While the current platform employs LSTM and GRU architectures, the rapidly evolving deep learning literature offers several promising architectural advances that CosmoPredict will explore in future iterations. Transformer-based sequence models — which have demonstrated state-of-the-art performance on time-series forecasting tasks in multiple domains — offer the potential for superior long-range dependency capture compared to recurrent architectures. Vision Transformer (ViT) models applied to solar disk imagery from SDO could enable direct flare prediction from solar magnetogram features.

6.2.4 Mobile Application and API Access

To extend CosmoPredict's operational reach beyond desktop-based mission control environments, a dedicated mobile application for iOS and Android platforms is planned for future development. The mobile application will deliver push notifications for G3+ storm alerts, provide a streamlined dashboard view optimised for mobile screen dimensions, and enable satellite operators to review and approve Route Diversion Module recommendations from the field. Additionally, a public REST API will be developed to allow third-party integration — enabling commercial satellite operators, insurance companies, and research institutions to programmatically access CosmoPredict's prediction outputs and incorporate them into their own operational systems.

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CosmoPredict: An Advanced AI-Driven Framework for Real-Time Space Weather Forecasting and Satellite Operational Resilience

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Abstract— The rapid proliferation of satellite constellations in both low Earth orbit and geostationary orbits has transformed space weather forecasting into a critical operational necessity, superseding its status as a purely scientific pursuit. Solar flares, coronal mass ejections, solar energetic particle events, and geomagnetic storms possess the capacity to disrupt spacecraft electronics, increase atmospheric drag, degrade communication links, and jeopardize ground infrastructure dependent on satellite services.

This paper talks about CosmoPredict, a single platform for predicting space weather in real time and making satellites more resilient. NASA OMNIWeb, NOAA Space Weather Prediction Center resources, and ACE-derived solar wind measurements all work together in the framework. To model how geomagnetic indicators like Kp and Dst change over time, we use deep learning architectures, especially Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks.

The system has a live monitoring dashboard, a three-level alert classification layer, and an Orbital Route Diversion Module that turns forecasts into actions for operators. Testing shows that the selected forecasting task has a predictive accuracy of 94.7%, with an RMSE of 0.042 and an MAE of 0.031. The suggested framework shows how machine learning can connect observational space science with safety planning for missions.

Keywords— Space Weather Forecasting; LSTM, GRU, Satellite Resilience, Geomagnetic Storm Prediction, CosmoPredict, Deep Learning, Orbital Route Diversion.

I. INTRODUCTION

Space-based infrastructure now supports navigation, communication, climate observation, remote sensing, strategic surveillance, disaster response, and modern financial synchronization. Any major drop in satellite performance can have effects that go far beyond the spacecraft itself. The orbital environment is also always at risk from a variety of dangerous solar events. Changes in the solar wind, magnetic field, coronal mass ejections (CMEs), and particle storms can change the radiation conditions around spacecraft and cause strong geomagnetic storms near Earth. As launches of satellite become more frequent, the costs associated with alarm systems that provide neither timely nor accurate information are also increasing.

Current space weather procedures heavily rely on observational portals, manual interpretation, alerts based on set thresholds, and domain experts who translate scientific data into operational decisions. While these systems are useful for monitoring, they often don't provide decision support that is predictive, integrated, and focused on taking action. Therefore, operators need a system that can handle incomplete data, forecast upcoming events, clearly explain risks, and suggest risk-reduction actions with enough time to be useful in practice.

This is why CosmoPredict was created. The project brings together a data ingestion pipeline, temporal deep learning models, risk-level classification, and decision-support logic into one working platform. Instead of treating forecasting as a separate analytics task, the system includes it as a step in a bigger operational workflow. A dashboard combines, processes, and displays historical and real-time indicators so that users can see how serious, how the trend is going, and what the likely effects of space weather events will be.

Another reason for this work is that machine learning is becoming more important for making scientific predictions. The growing number of publicly available datasets from NASA and NOAA has made it possible to use recurrent models on patterns that were previously handled with rule-based or specialized methods. Recurrent neural networks, especially LSTM and GRU architectures, work well with sequential data where the order of the data is important. They can learn delayed correlations, cyclic behavior, and precursor patterns that might not be obvious when you look at them by hand.

The main goal is to make space weather forecasting understandable at the system level. A prediction is only useful in practice if it helps us to make a specific choice. This work goes far beyond just accuracy metrics. It also includes an alerting layer and an Orbital Route Diversion Module.

Divyansh Srivastava

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PROFILE

Python Full Stack Developer with hands-on experience in building scalable web applications using **Django**, **Django REST Framework**, **React.js**, and **PostgreSQL**. Strong understanding of backend development, RESTful APIs, and responsive frontend design. Skilled in writing clean, efficient code and solving real-world problems, with experience delivering reliable and user-focused solutions.

TECHNICAL SKILLS

Languages: C, Python, JavaScript

Development Skills: React, Django, Django REST Framework

ML Frameworks / Libraries: NumPy, Pandas, Matplotlib, Scikit-Learn

Databases: SQL, PostgreSQL, Firebase

Cloud & DevOps: AWS, Docker, Nginx

Tools: Git, Postman, Excel, PowerPoint, VS Code, IntelliJ IDEA, Google Colab

Soft Skills: Report Building, Stakeholder Management, People Management

EXPERIENCE

Metagates Innovation Pvt. Ltd.

India

Software Development Intern

Mar 2025 – Aug 2025

- Developed interactive and responsive frontend components using JavaScript and modern UI practices, improving usability and performance.
- Designed and deployed an intelligent chatbot interface for **Nakula**, reducing query resolution time by **40%**.
- Implemented **8+ dynamic and SEO-friendly web pages**, boosting user engagement by **65%**.

Mittal Alliance

India

Web Development Intern

Oct 2025 – Nov 2025

- Built mobile-first, responsive UI layouts using HTML, CSS, and JavaScript.
- Collaborated with backend teams to integrate APIs and optimize frontend performance.

PROJECTS

CosmoPredict 🔗 | Full Stack (Django, React.js, ML)

Major Project

- Developed a full stack predictive platform using Django, React.js, and machine learning models to deliver personalized insights, integrating RESTful APIs, dynamic UI, and data-driven features for enhanced user experience.

Vedique Preschool Website 🔗 | Full Stack (React.js, Firebase)

Freelance

- Designed and developed a fully responsive full stack preschool website using React.js and Firebase, implementing real-time backend services, authentication, and optimized performance with a user-friendly UI.

Collex 360 Website 🔗 | Django, Machine Learning

- Developed a student companion platform using Django and machine learning to manage attendance, expenses, health, and skills in one place, based on research of student needs and existing solutions.

EDUCATION

Institute of Technology and Management, GIDA

Gorakhpur, India

B.Tech in Computer Science (AI & ML) | GPA: 8.5

Sep 2022 – Jun 2026

SVM Inter College, Arya Nagar

Gorakhpur, India

Intermediate | Percentage: 82.6

Apr 2021 – Mar 2022

AWARDS AND ACHIEVEMENTS

- **Smart India Hackathon 2024 (Runner-Up)** — Secured Runner-Up in Smart India Hackathon 2024
- **HackSetu 1.0 Hackathon** — Participated in HackSetu 1.0 National Level Hackathon and got **4th Rank** among all the participating teams.
- **Head – College Coding Committee:** Organized coding and web-development contests during annual Tech Fest.

KRISHNA KANT SHARMA

Generative AI & Machine Learning Engineer

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SUMMARY

Generative AI and Machine Learning practitioner with hands-on experience building and deploying production-grade AI systems. Specialized in LLM-powered agent architectures, natural language interfaces, and end-to-end deployment pipelines. Proficient in LangChain, LangGraph, Groq, and prompt engineering for real-world applications. Complementary background in classical ML workflows including supervised learning, data preprocessing, and model evaluation using Python and Scikit-learn.

SKILLS

Programming & Querying: Python, SQL

Generative AI: Large Language Models (LLMs), LangChain, LangGraph, Groq, Prompt Engineering, AI Agents, SQL Agents, RAG Fundamentals

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation

Deep Learning: Neural Networks, CNNs, TensorFlow, PyTorch

Data Analysis: Pandas, NumPy, Data Cleaning, Exploratory Data Analysis (EDA)

Database: PostgreSQL, SQLAlchemy, SQL

Deployment Tools: Streamlit, Supabase, Git, Jupyter Notebook

EXPERIENCE

Data Science and Machine Learning Intern

July 2025 – September 2025

YBI Foundation (Remote)

Tools Used: Python, Pandas, NumPy, Scikit-learn

- Completed a customer churn prediction project using supervised machine learning techniques.
- Performed data cleaning, exploratory data analysis (EDA), and feature engineering using Python.
- Trained and evaluated ML models using Scikit-learn and standard evaluation metrics.

PROJECTS

TaskBot Pro – AI-Powered Task Management System

March 2026

Python, LangChain, LangGraph, Groq, PostgreSQL, Supabase, Streamlit

- Architected a LangChain SQL agent powered by Groq's LLM with SQLDatabaseToolkit, enabling dynamic natural language to SQL query generation and execution against a live PostgreSQL database. Architected a LangChain SQL agent powered by Groq's LLM with SQLDatabaseToolkit, enabling dynamic query generation, execution, and result interpretation from natural language prompts.
- Implemented multi-turn conversational memory using LangGraph's InMemorySaver and designed a multi-tenant isolation architecture with user-level row filtering enforced through LLM system prompt engineering. Designed a multi-tenant data isolation architecture with user-level PostgreSQL row filtering enforced through LLM system prompt engineering — ensuring zero data leakage between users.
- Built persistent custom authentication backed by Supabase PostgreSQL with bcrypt-hashed credentials, replacing file-based auth that wiped on every redeploy.
- Deployed on Streamlit Cloud with Supabase Session Pooler, SSL enforcement, and NullPool for production-grade cloud database connectivity.

Ask Buddy – Conversational AI Q&A Chatbot

January 2026

Python, LangChain, Groq (LLaMA 3.1), Streamlit

- Built and deployed a multi-turn conversational AI chatbot using LangChain with Groq-hosted LLaMA 3.1.
- Implemented session-based conversation memory to maintain context across user interactions.
- Designed dynamic prompt construction using chat history to improve response relevance.
- Deployed with secure API key management via environment variables on Streamlit.

CERTIFICATIONS

- Claude Code in Action – Anthropic (2026)
- Generative AI Program – Outskill (2025)

EDUCATION

B.Tech – Computer Science & Engineering (AI & ML)

2022 – 2026

Institute of Technology and Management, GIDA, Gorakhpur

7.06 CGPA

Ajay Kumar

Gorakhpur, U.P | ajayadav7376@gmail.com | 7355979473 | linkedin.com/in/ajayadav7769
github.com/ajayadav-7769

Objective

Passionate AI/ML enthusiast and creative problem-solver in my final year of B.Tech in Artificial Intelligence and Machine Learning. at the Institute of Technology and Management. I am actively seeking an opportunity to learn from industry leaders, grow my technical expertise, and make a tangible impact on cutting-edge projects.

Education

Institute Of Technology And Management, GIDA, B.tech in Artificial Intelligence and Machine Learning Sept 2022 – May 2026

- GPA:6.5

National Institute of Electronics and Information Technology,O-Level Jan 2026

Technologies

Languages: C, JavaScript, SQL, Python

Tools: Excel, PowerPoint, MySQL , Postman , PowerBI

Frameworks and Libraries: Matplotlib, Scikit-Learn, NumPy, Pandas, Tailwind CSS

Platforms: Visual Studio Code, Google Collab , Jupyter Notebook

Soft Skills:Communication, Problem-Solving, Team Collaboration, Adaptability

Experience

MERN STACK (intern), AppWorks Technologies Pvt. Ltd.(Lucknow) June 2025- Aug 2025

- Contributed to the end-to-end development of a full-stack web application using the MERN (MongoDB, Express.js, React.js, Node.js) stack.
- Identified, debugged, and resolved software defects, improving application stability and user experience.
- Designed and implemented MongoDB schemas with Mongoose to model application data effectively for features like user profiles

Web Developer (intern),Oasis Infobyte Nov 2024 – Dec 2024

- Developed responsive web pages using HTML5 and CSS3, ensuring cross-browser compatibility and mobile optimization.
- Implemented custom styling for buttons, forms, and navigation menus to enhance user experience.

Projects

MERN Stack User Authentication System

- Developed a full-stack web application featuring secure user registration and login functionality.
- Utilized MongoDB for user data storage and Postman for API endpoint testing.
- Implemented password hashing with bcrypt and session management using JSON Web Tokens (JWT) for enhanced security.

DEV'S NOTES

- Developed DEV'S NOTES, a static website featuring a digital E-Book library and study resources, using HTML for content structure.
- Designed and implemented a clean, user-friendly, and fully responsive layout with CSS, ensuring optimal viewing on all devices.

SHANTANU SHAHI

Bachelor of Technology
Dr. A.P.J. Abdul Kalam Technical University, Lucknow UP

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shantanushahi9@gmail.com
GitHub Profile
LinkedIn Profile

EDUCATION

- **Bachelor of Technology in Artificial Intelligence and Machine Learning** 2022-26
Institute of Technology and Management, Gorakhpur Percentage: 66
2022-26

PROJECTS

- **English-Sanskrit Bidirectional Translator**
World's most accurate English-Sanskrit translator based on VEDAS
 - End to end Translation
 - Using NLP for Translation
 - **Technology Used:** Python, TensorFlow, NLTK
- **AI-Powered Sports Motion Analysis Tool**
AI-powered system that analyzes the cricket cover-drive technique using computer vision and ML to give real time feedback.
 - Real-time pose tracking using Media-pipe + OpenCV.
 - Shot accuracy classification using TensorFlow/Keras.
 - Instant feedback system built with Python and ML model

EXPERIENCE

- **AIML using Python Internship** JULY - AUGUST 2025
National Institute of Electronics and Information Technology (NIELIT), Gorakhpur. Offline
 - Scientific Computing Foundation: Mastery of NumPy for high-performance vectorized operations using Nd arrays (N- dimensional arrays).
 - Model Implementation: Hands-on experience with the Scikit-learn (Sklearn) library to fit and predict using models like k-NN and Support Vector Machines (SVM)
 - Data Wrangling: Practical skills in data munging using Pandas, including missing values imputation and Feature Engineering (e.g., One-Hot Encoding).
- **Machine Learning Internship** JUN-AUG 2025
Internshala-Scholiverse Educare Pvt. Ltd Online
 - Linear Regression, Decision Trees) using Python.
 - Decision Tree, Information Gain, Ensemble Methods
 - Confusion Matrix, False Positives, Cross-Validation
 - Exploratory Data Analysis (EDA), Feature Importance, Python Libraries
- **Innovation Ambassador Training** APRIL - AUGUST 2025
AICTE (All India Council of Technical Education) (MOE), Delhi. Online
 - Foundation Level:(Design Thinking, Ideation, Needs Assessment).
 - Advance Level:(Innovation Management, IPR, Minimum Viable Product (MVP))
 - Re-skilling Level:(Technological Trends, Disruptive Technologies, Business Modeling)

TECHNICAL SKILLS AND INTERESTS

Languages: C/C++, Python, JavaScript, HTML+CSS

Libraries: C++ STL, Python Libraries, ReactJs **Web**

Dev Tools: Nodejs, VScode, Git, GitHub **Frameworks:** ReactJs

Cloud/Databases: MongoDB, Firebase, Relational Database (MySQL)

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object Oriented Programming, Database Management System, Software Engineering.

Areas of Interest: Web Design and Development, Cloud Security.

Soft Skills: Problem Solving, Self-learning, Presentation, Adaptability

CERTIFICATION AND AWARDS

- **Smart India Hackathon2024 (Themed AIR-2)** Amity University, Chhattisgarh Nov-2024
- **National Entrepreneurship Challenge 2025 (AIR-124)** IIT Bombay, Mumbai Sept-2025
- **National Digital Literacy Mission 2015** MCIT-Government of India Oct-2025